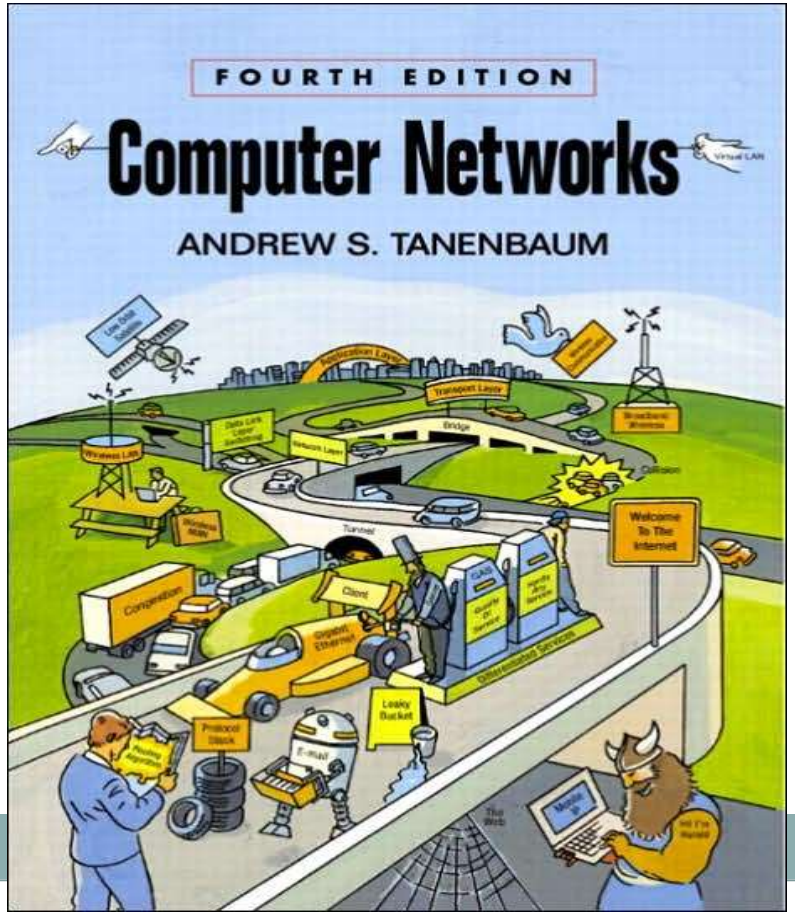
Computer Networks

Textbook:

Computer Networks 4th ed., by A.S. Tanenbaum, 新月



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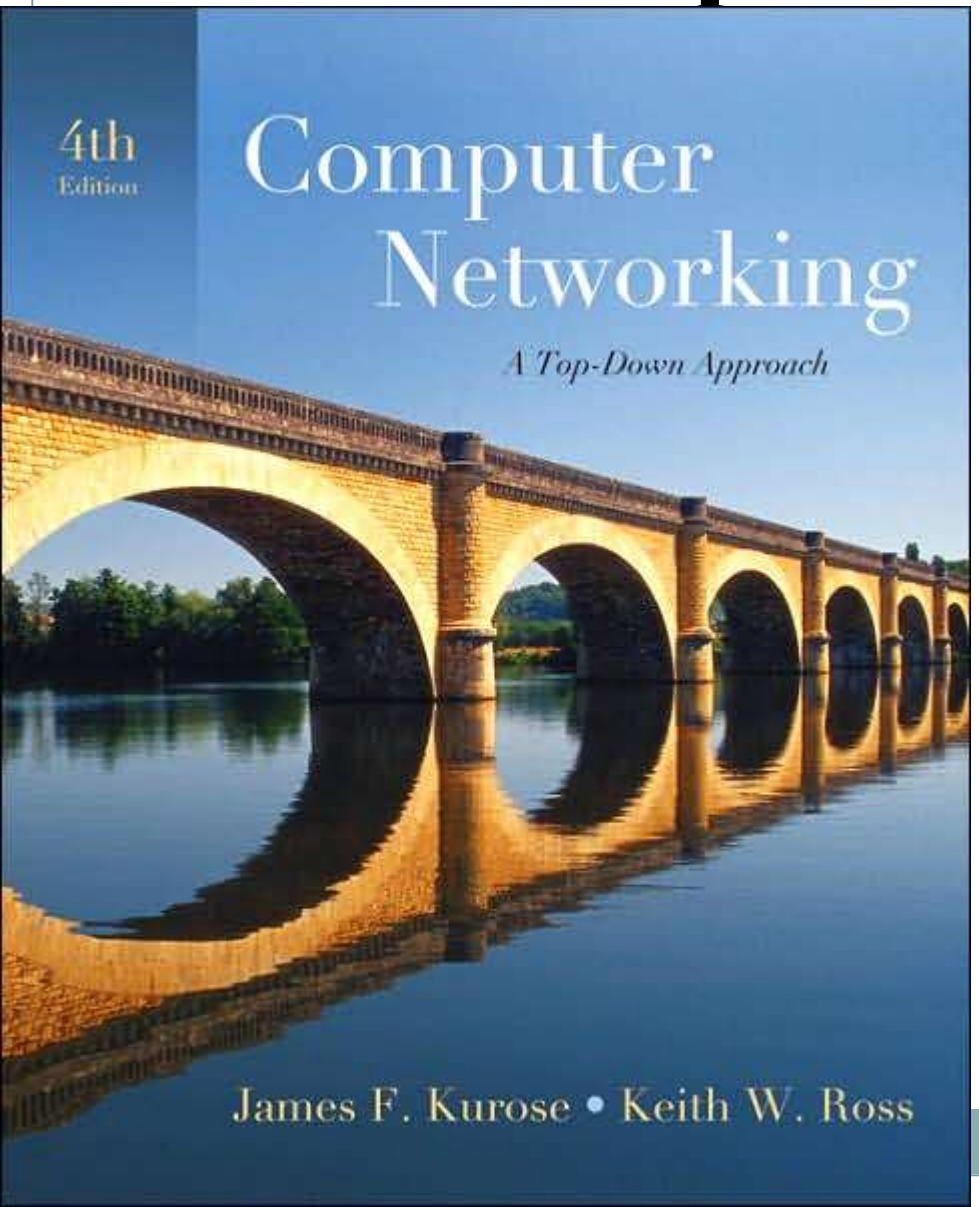
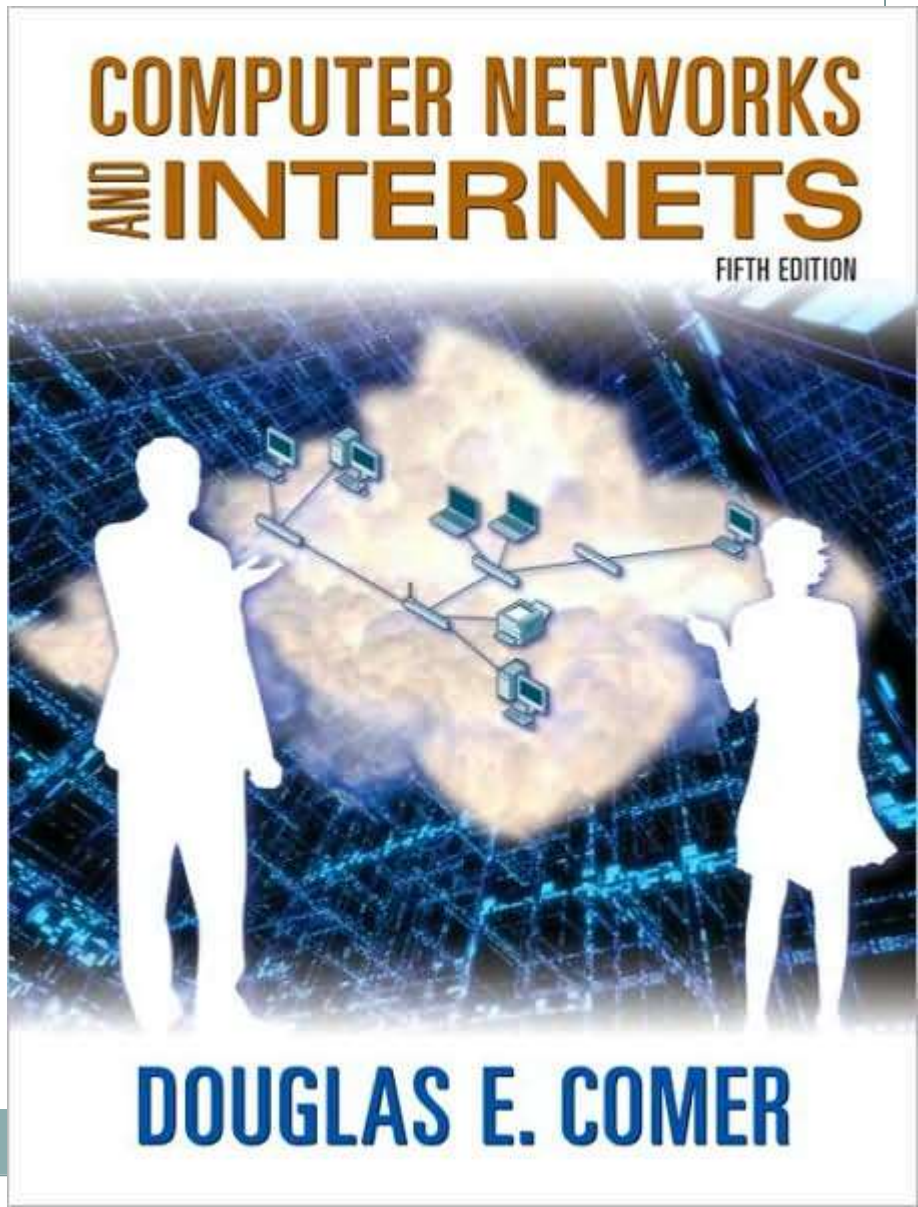




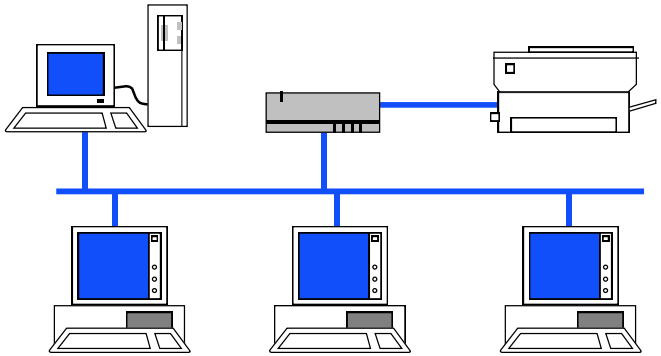
Computer Networks

Course Contents:

Chapter 1 INTRODUCTION
Chapter 2 THE PHYSICAL LAYER
Chapter 3 THE DATA LINK LAYER
Chapter 4 THE MEDIUM ACCESS SUBLAYER
Chapter 5 THE NETWORK LAYER
Chapter 6 THE TRANSPORT LAYER

Chapter 1: Introduction

Computer Networks vs. Distributed Systems



Users make the decision.

O.S. makes the decision.

Chapter 1: Introduction

Middleware: The software layer that lies between the operating system and applications on each side of a distributed computing system in a network.

Middleware services provide a more functional set of application programming interfaces to allow an application to:

1. Locate transparently across the network, thus providing interaction with another service or application
2. Be independent from network services
3. Be reliable and available always

when compared to the operating system and network services.

Chapter 1: Introduction

1.1 Uses of Computer Networks

1.1.1 Business Applications

Goals:

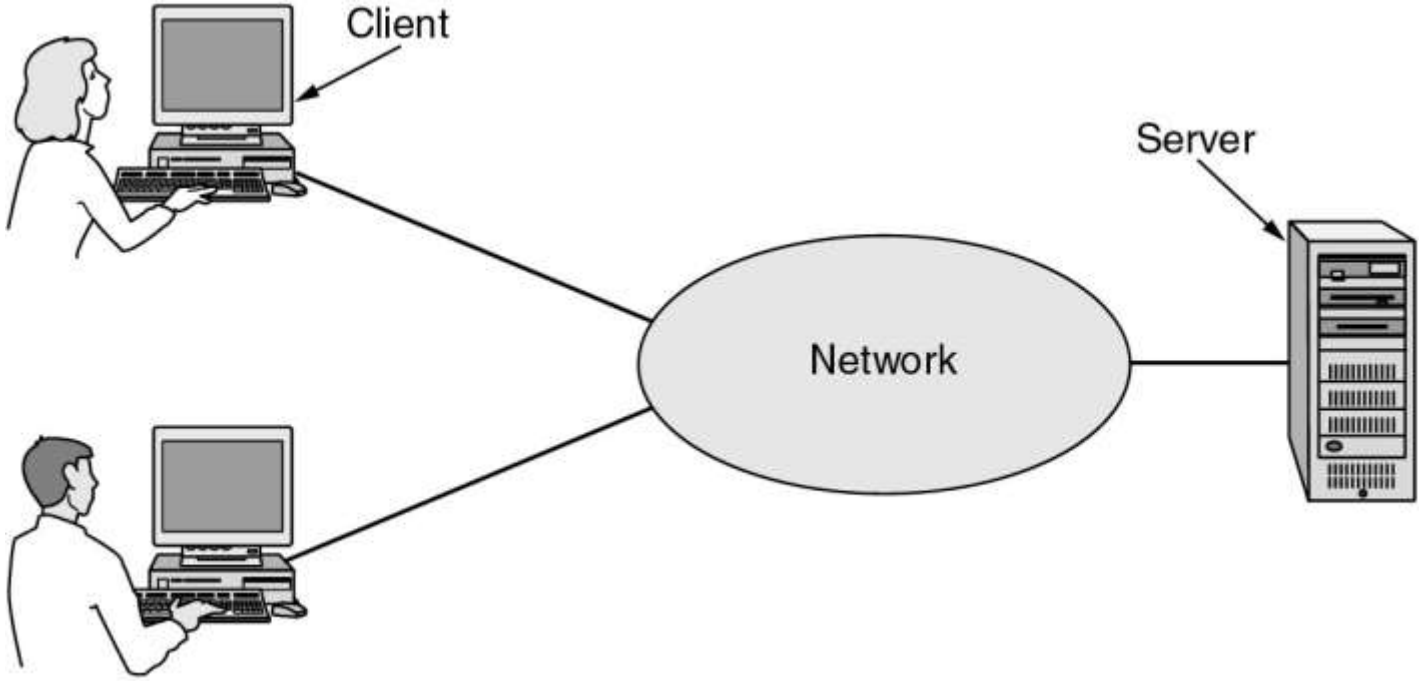
- Resource Sharing
- High Reliability
- Saving Money
- Communication Medium (people to people, machine to machine)

Chapter 1: Introduction

1.1 Uses of Computer Networks

1.1.1 Business Applications

Client-server model

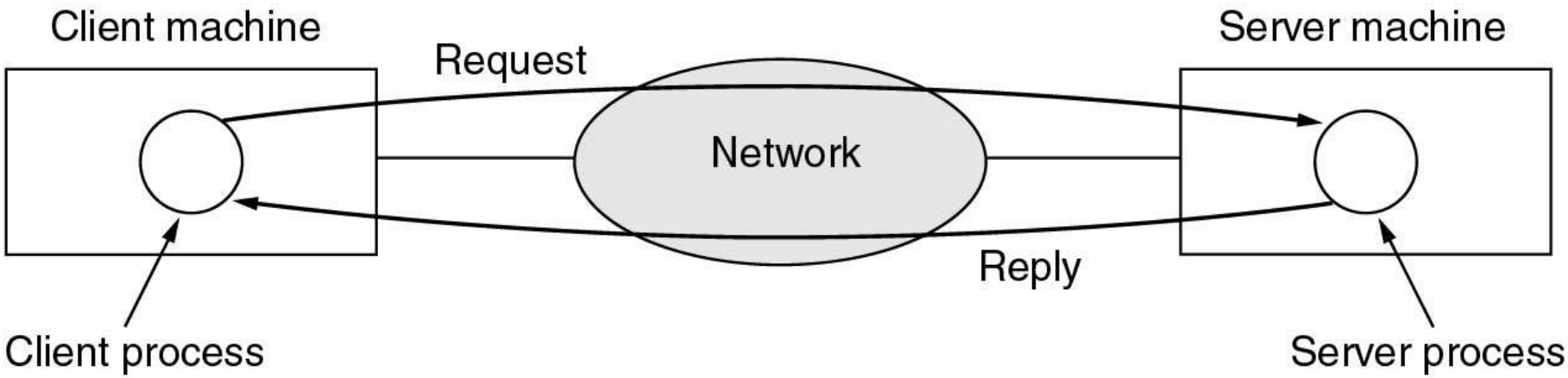


A network with two clients and one server.

Chapter 1: Introduction

1.1 Uses of Computer Networks

1.1.1 Business Applications



The client-server model involves requests and replies.

Chapter 1: Introduction

1.1 Uses of Computer Networks

1.1.1 Business Applications

Client-server model

Concurrent server vs. Iterative server

Stateful server vs. Stateless server

Examples:

Chapter 1: Introduction

1.1 Uses of Computer Networks

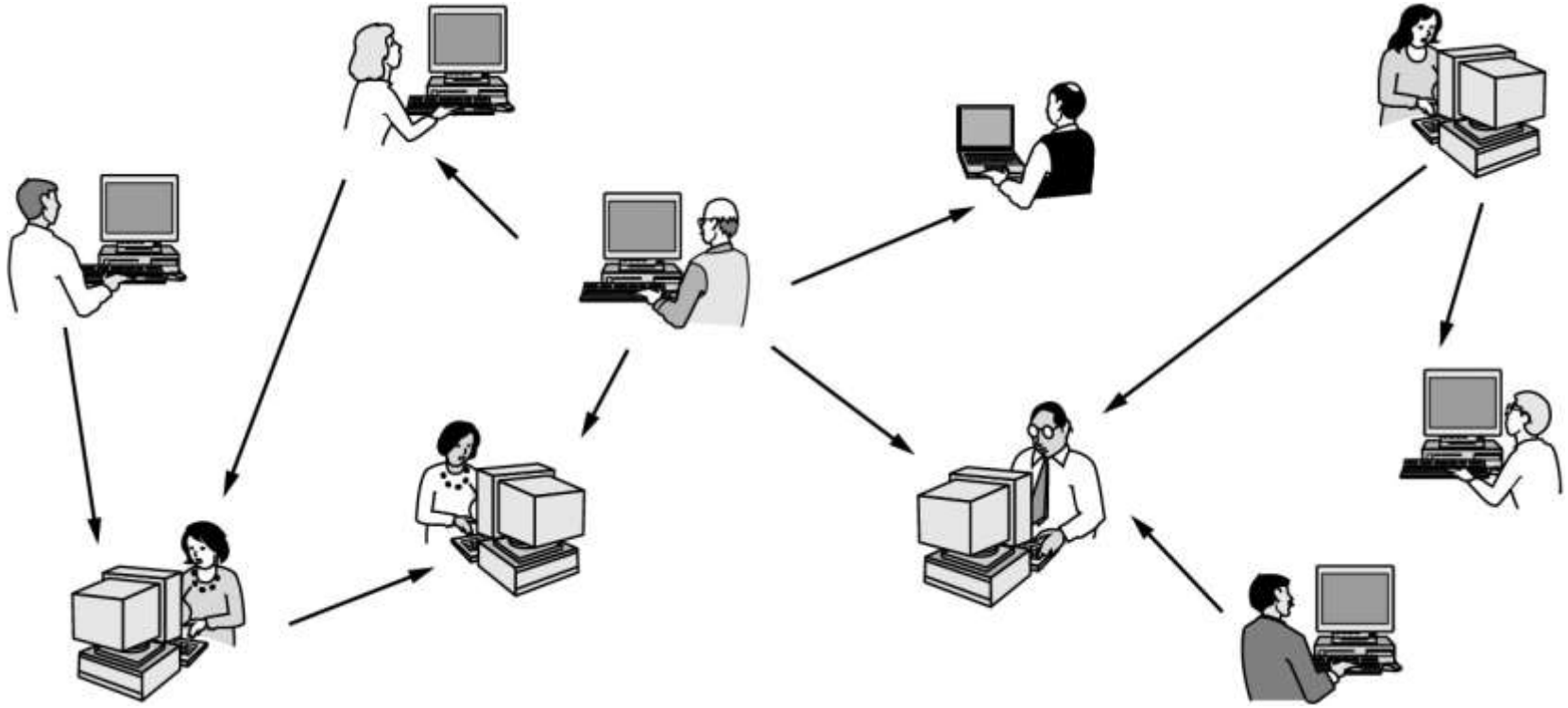
1.1.2 Home Applications

- Access to remote information
  - World Wide Web
- Person-to-person communication
  - Electronic mail, IM, Videoconference
- Interactive entertainment
  - Video-on-Demand, Games
- Electronic commerce

Chapter 1: Introduction

1.1 Uses of Computer Networks

1.1.2 Home Applications



In peer-to-peer system there are no fixed clients and servers.

Chapter 1: Introduction

1.1 Uses of Computer Networks

1.1.2 Home Applications

Tag	Full name	Example
B2C	Business-to-consumer	Ordering books on-line
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products on-line
P2P	Peer-to-peer	File sharing

Some forms of e-commerce.

Chapter 1: Introduction

1.1 Uses of Computer Networks

1.1.3 Mobile Users

Wireless	Mobile	Applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	Yes	Portable office; PDA for store inventory

Combinations of wireless networks and mobile computing.

Chapter 1: Introduction

1.1 Uses of Computer Networks

1.1.3 Mobile Users

Cellular Phones

PDA: Personal Digital Assistant

Smart Phones: PDA+Handset+GPS+…

Chapter 1: Introduction

1.1 Uses of Computer Networks

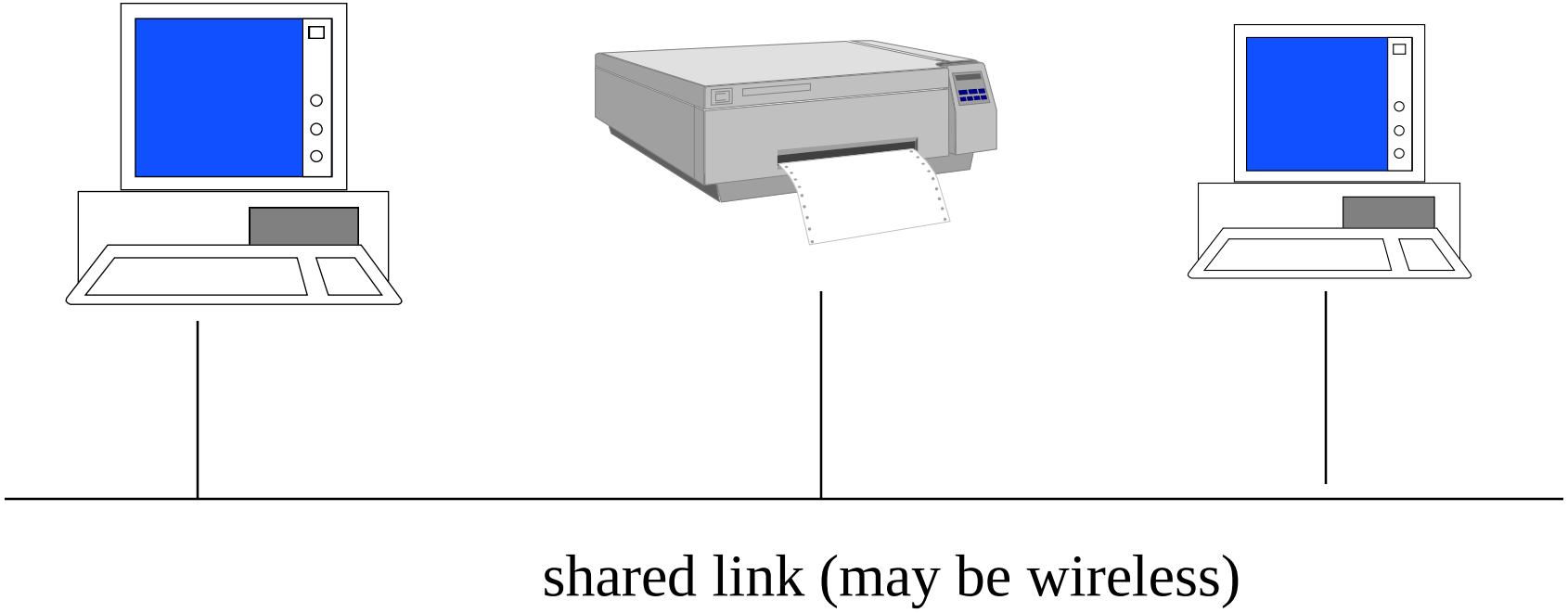
1.1.4 Social Issues

Privacy
Copyright
Anonymity
Security
Worms and Virus
freedom of speech vs. censorship
responsibility of the service providers
…

Chapter 1: Introduction

1.2 Network Hardware

Broadcast Networks



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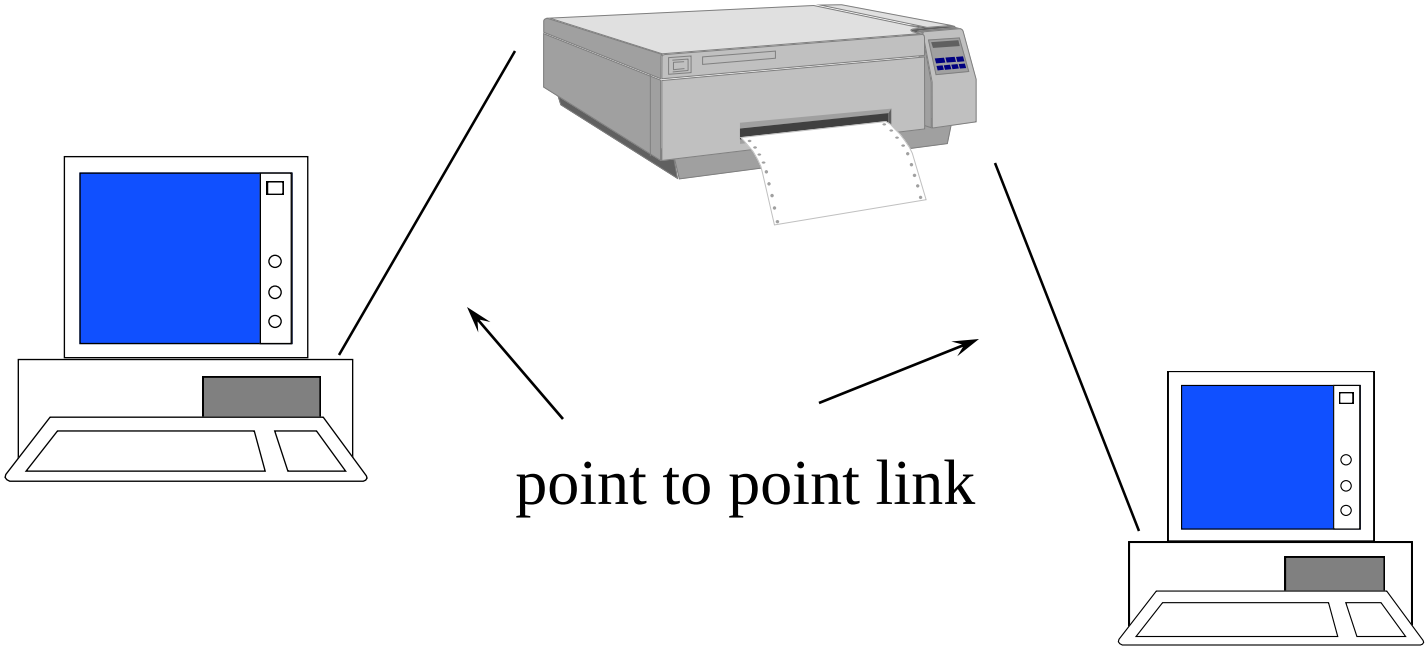
1.2 Network Hardware

Point-to-point Networks



Chapter 1: Introduction

1.2 Network Hardware

Classifying by scales

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Classification of interconnected processors by scale.

Chapter 1: Introduction

1.2 Network Hardware

1.2.1 Local Area Networks

IEEE的定義：

A LAN (Local Area Network) is a ***data communication system*** allowing a number of ***independent devices*** to communicate directly with each other, within a ***moderately sized geographic area*** over a physical communication channel of ***moderate data rates***.

Chapter 1: Introduction

1.2 Network Hardware

1.2.1 Local Area Networks

LAN:

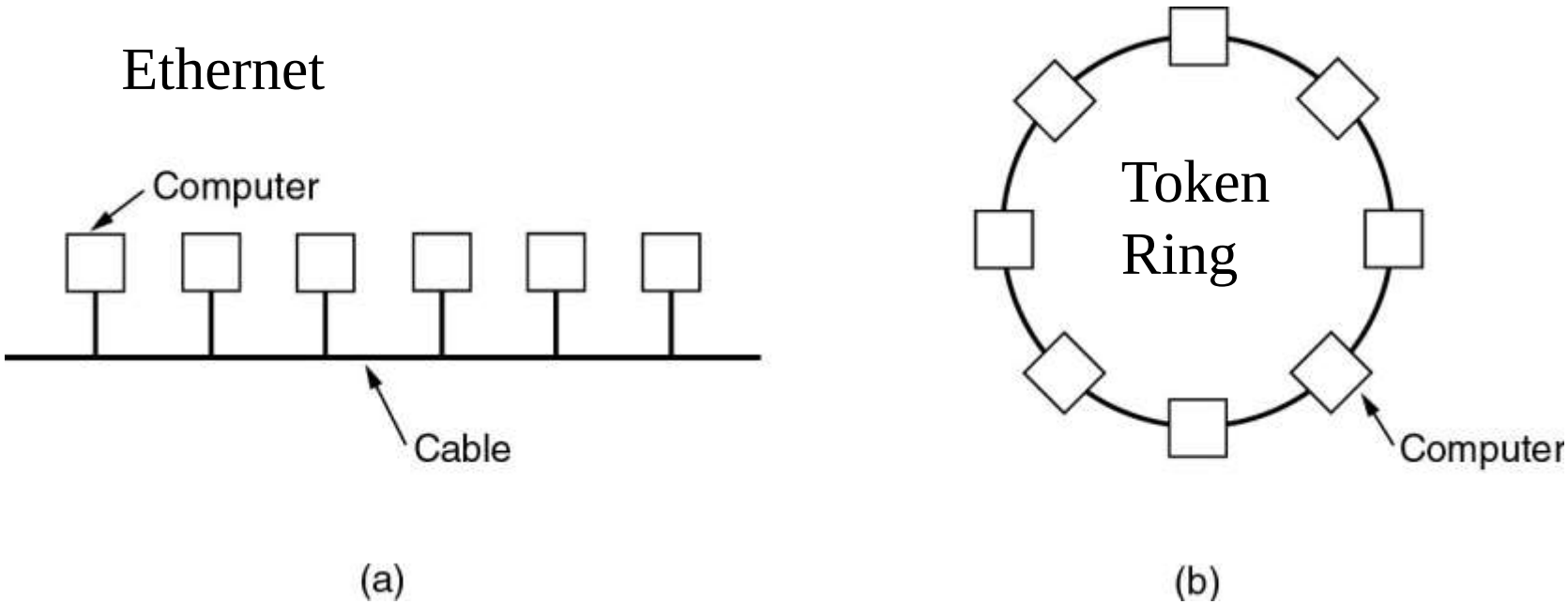
- short geographical distance (a few kilometers)
- high speed (Larger than 1 Mbps)
- multiple access (Many can use it at the same time)
- sharing (hardware, software, idea,...)

Chapter 1: Introduction

1.2 Network Hardware

1.2.1 Local Area Networks

Ethernet



Two broadcast networks
(a) Bus
(b) Ring

Chapter 1: Introduction

1.2 Network Hardware

1.2.1 Local Area Networks

Standardization Body

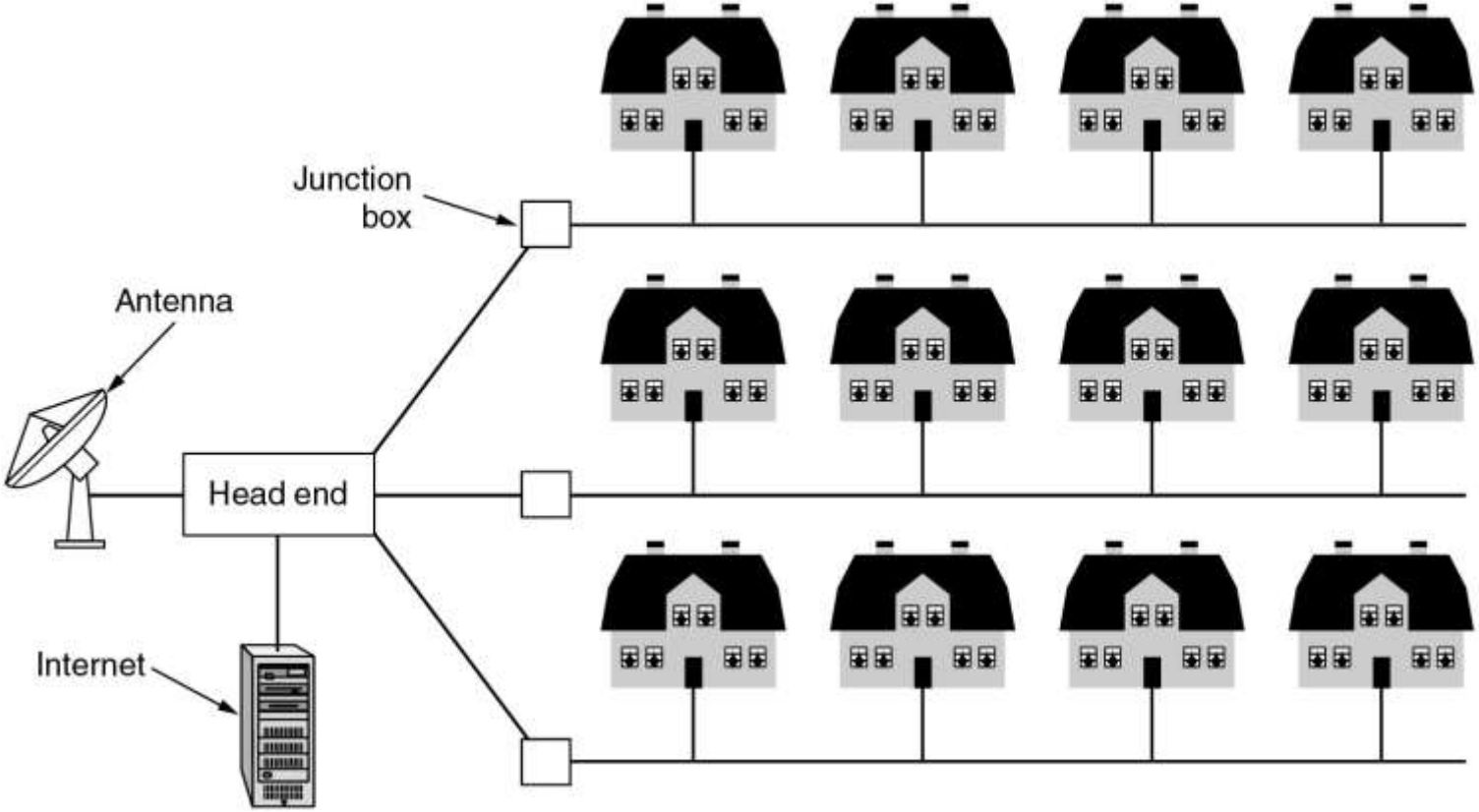
IEEE (Institute of Electric and Electronic Engineers) 802 group

For example:
802.3: CSMA/CD (Carrier Sense Multiple Access with Collision Detection) (Ethernet is one of them.)
802.11: Wireless LANs (Wi-Fi: Wireless Fidelity)

Chapter 1: Introduction

1.2 Network Hardware

1.2.2 Metropolitan Area Networks



A metropolitan area network based on cable TV.

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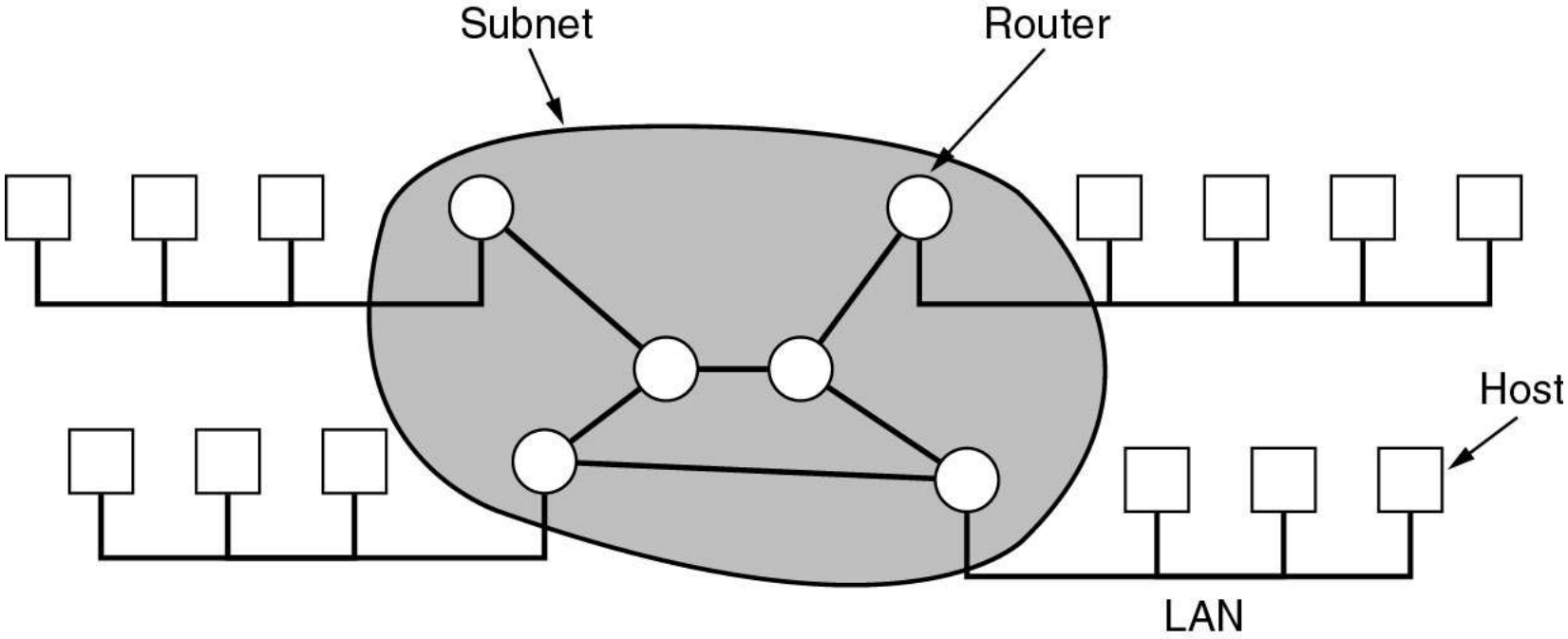
1.2 Network Hardware

1.2.3 Wide Area Networks



Relation between hosts on LANs and the subnet.

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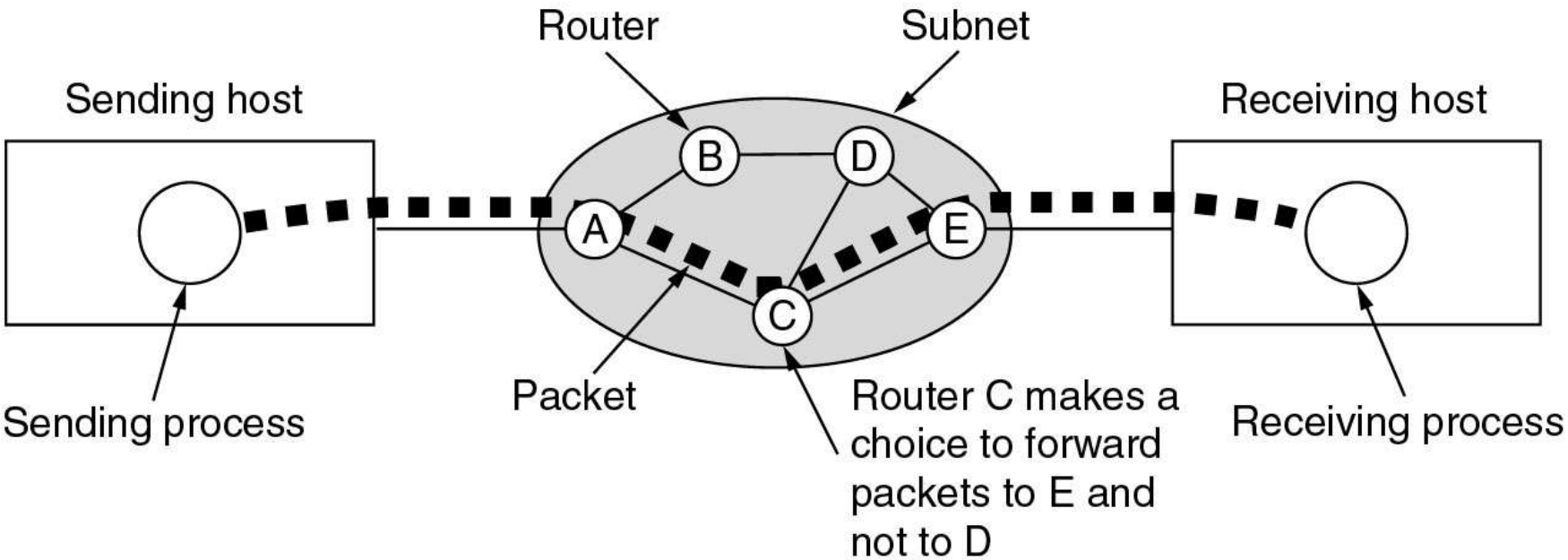
1.2 Network Hardware

1.2.3 Wide Area Networks



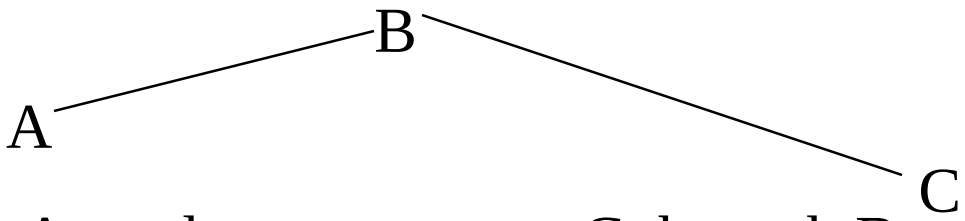
A stream of packets from sender to receiver.

Chapter 1: Introduction

1.2 Network Hardware

1.2.3 Wide Area Networks

store-and-forward network



A sends a message to C through B.

B must store this message until B is sure that C has received it.

Store first, then forward. But when to start forwarding?

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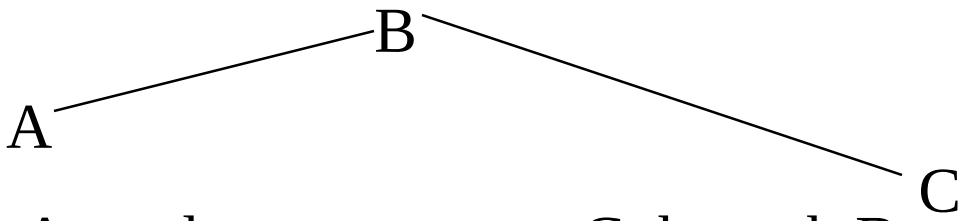
1.2 Network Hardware

1.2.3 Wide Area Networks

store-and-forward network



A sends a message to C through B.

When to starting forwarding?

1. After the message is completely received
2. Start forwarding after a fixed amount of information(bits) received
3. Start forwarding immediately after receiving data (cut-through)

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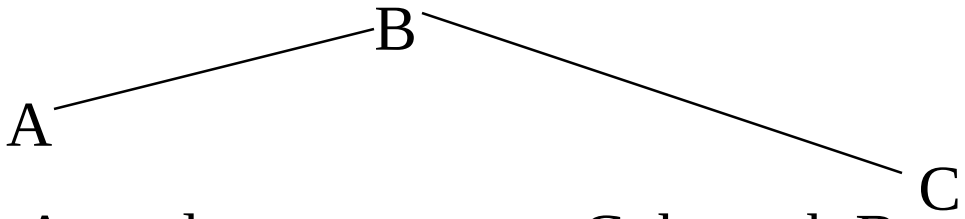
1.2 Network Hardware

1.2.3 Wide Area Networks

store-and-forward network



A sends a message to C through B.

	A	B	C
0	m1		
0.25	m2	m1	
0.5	m3	m2	m1
0.75	m4	m3	m2
1.0		m4	m3
1.25			m4

If a message takes 1 minute to travel a link:

(1) A to B, then B to A: 2 minutes

(2) message is decomposed into 4 parts: 1.25 minutes
(each part is called a packet)

Chapter 1: Introduction

1.2 Network Hardware

1.2.3 Wide Area Networks

The concept of pipelining

If the message is decomposed into N packets, each packet takes 1/N minutes to travel a link. It takes $\frac{2}{N} + \frac{N-1}{N}$ minutes.

bits

h		t
header	user information	trailer

overhead for N packets=$N(h+t)$

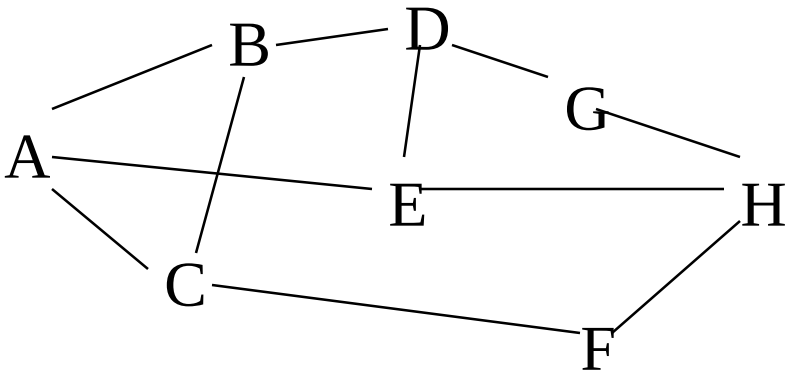
overhead for 1 message=$h+t$

Chapter 1: Introduction

1.2 Network Hardware

1.2.3 Wide Area Networks

Switching



(1) circuit switching (in telephone)
(2) packet switching
(3) message switching

Current Internet practice: store-and-forward packet switching

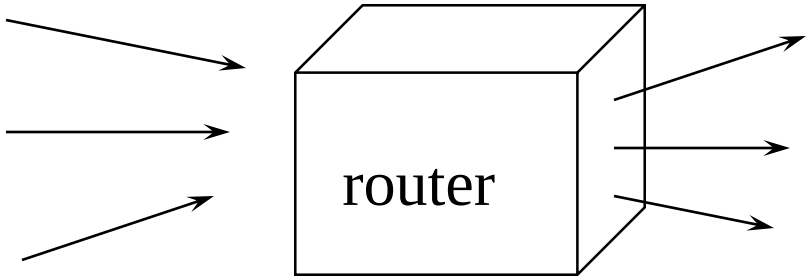
Chapter 1: Introduction

1.2 Network Hardware

1.2.3 Wide Area Networks

Switching

Bottleneck in current store-and-forward packet switching network:
ROUTING



packets delayed (or even discarded) in routers

Chapter 1: Introduction

1.2 Network Hardware

1.2.4 Wireless Networks

The fast-growing segment of the industry:
- notebook computers
- personal digital assistants
- cellular phones

Before long, we would have:
- palmtop computers
- wristwatch computers
- …

Chapter 1: Introduction

1.2 Network Hardware

1.2.4 Wireless Networks

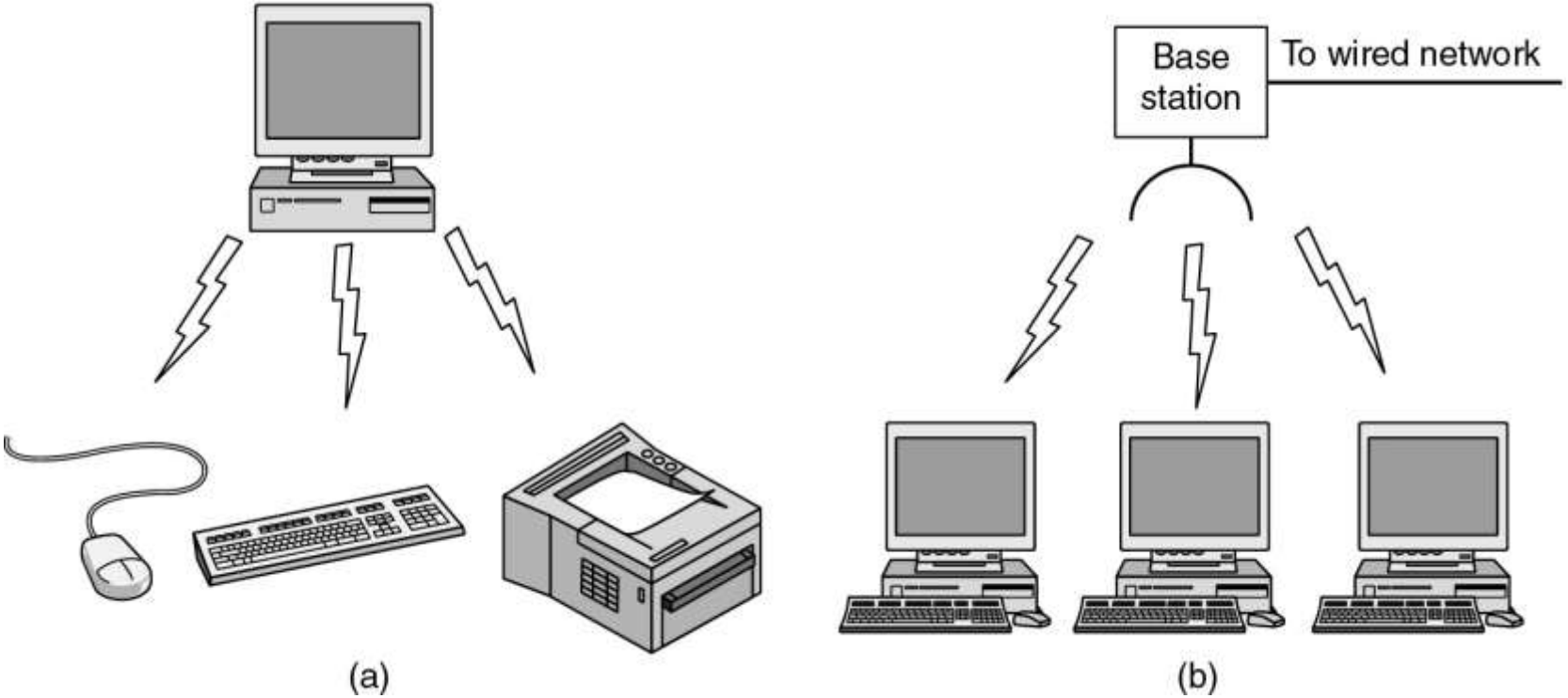
Categories of wireless networks:

1. (used for) System interconnection
2. Wireless LANs
3. Wireless WANs

Chapter 1: Introduction

1.2 Network Hardware

1.2.4 Wireless Networks



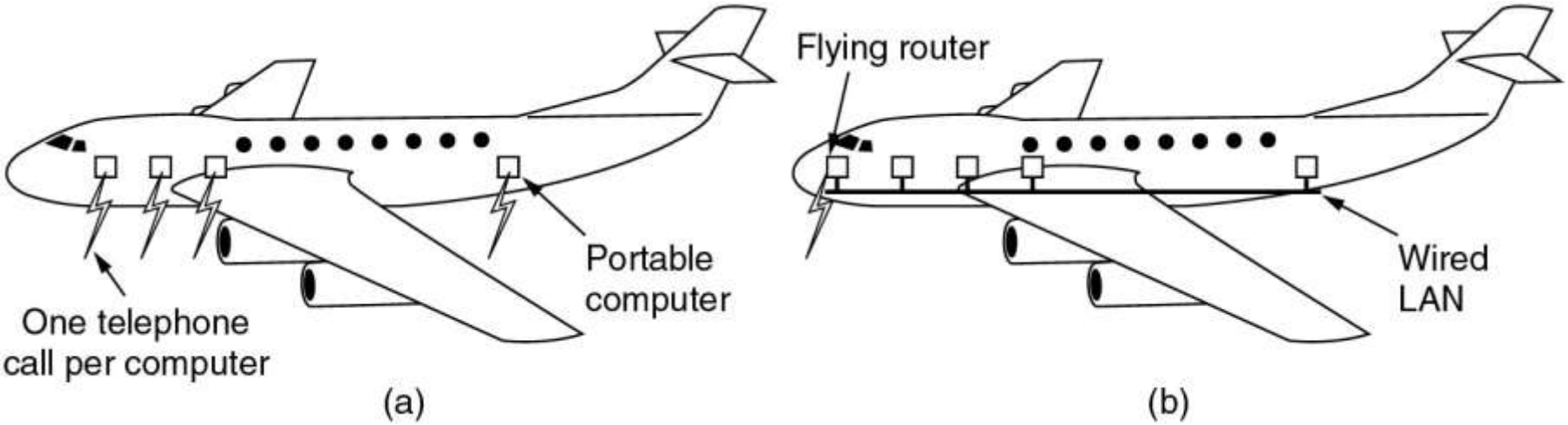
(a) Bluetooth configuration

(b) Wireless LAN

Chapter 1: Introduction

1.2 Network Hardware

1.2.4 Wireless Networks



(b) is more efficient and economical.

NEMO: NEtwork MObility

Chapter 1: Introduction

1.2 Network Hardware

1.2.5 Home Networks

1. Computers (desktop PC, PDA, shared peripherals
2. Entertainment (TV, DVD, VCR, camera, stereo, MP3)
3. Telecomm (telephone, cell phone, intercom, fax)
4. Appliances (microwave, fridge, clock, furnace, airco, lights)
5. Telemetry (utility meter, smoke/burglar alarm, thermostat, babycam).

Chapter 1: Introduction

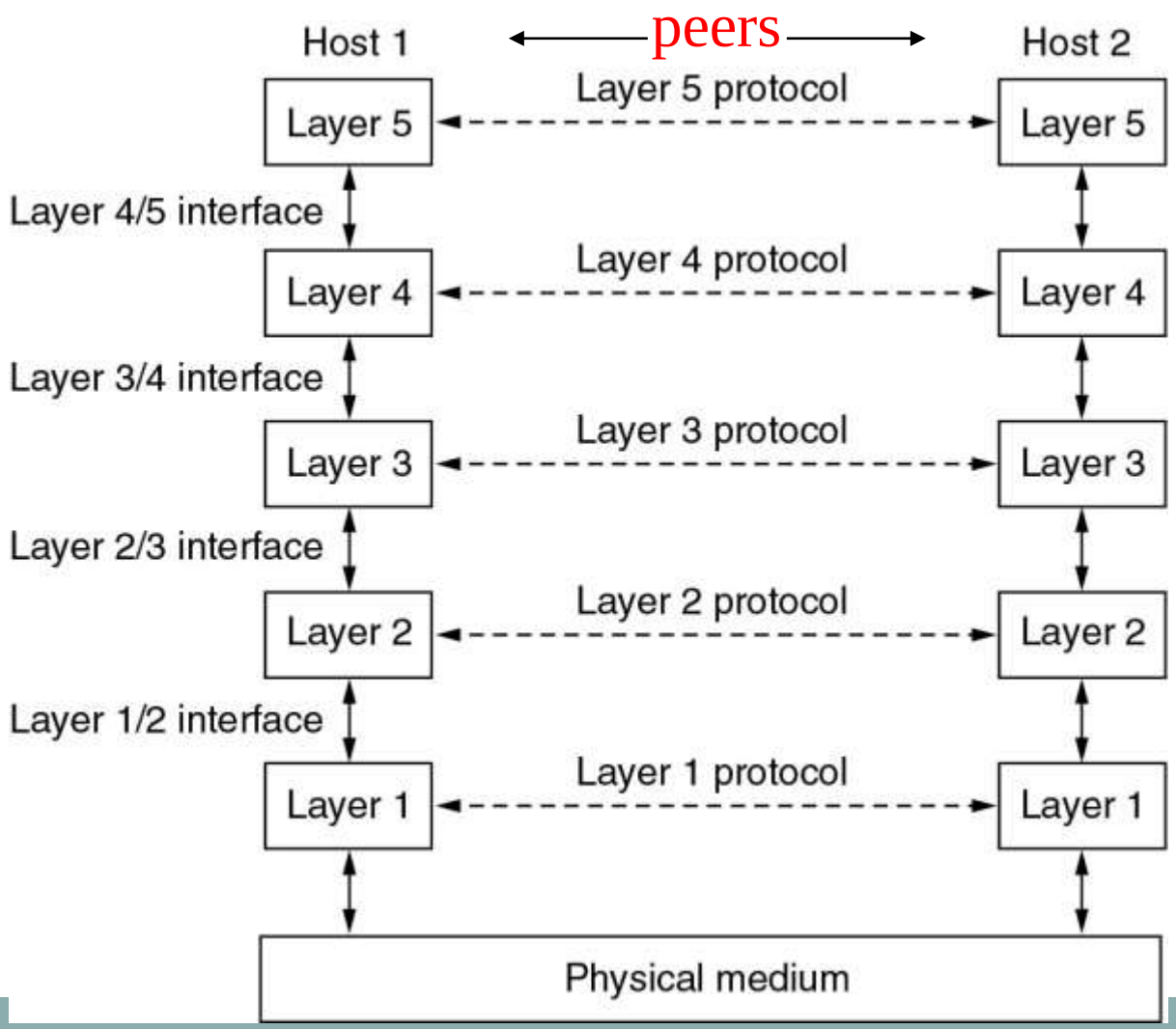
1.3 Network Software

1. Protocol Hierarchies
2. Design Issues for the Layers
3. Connection-Oriented and Connectionless Services
4. Service Primitives
5. The Relationship of Services to Protocols

Chapter 1: Introduction

1.3 Network Software

1.3.1 Protocol Hierarchies



Chapter 1: Introduction

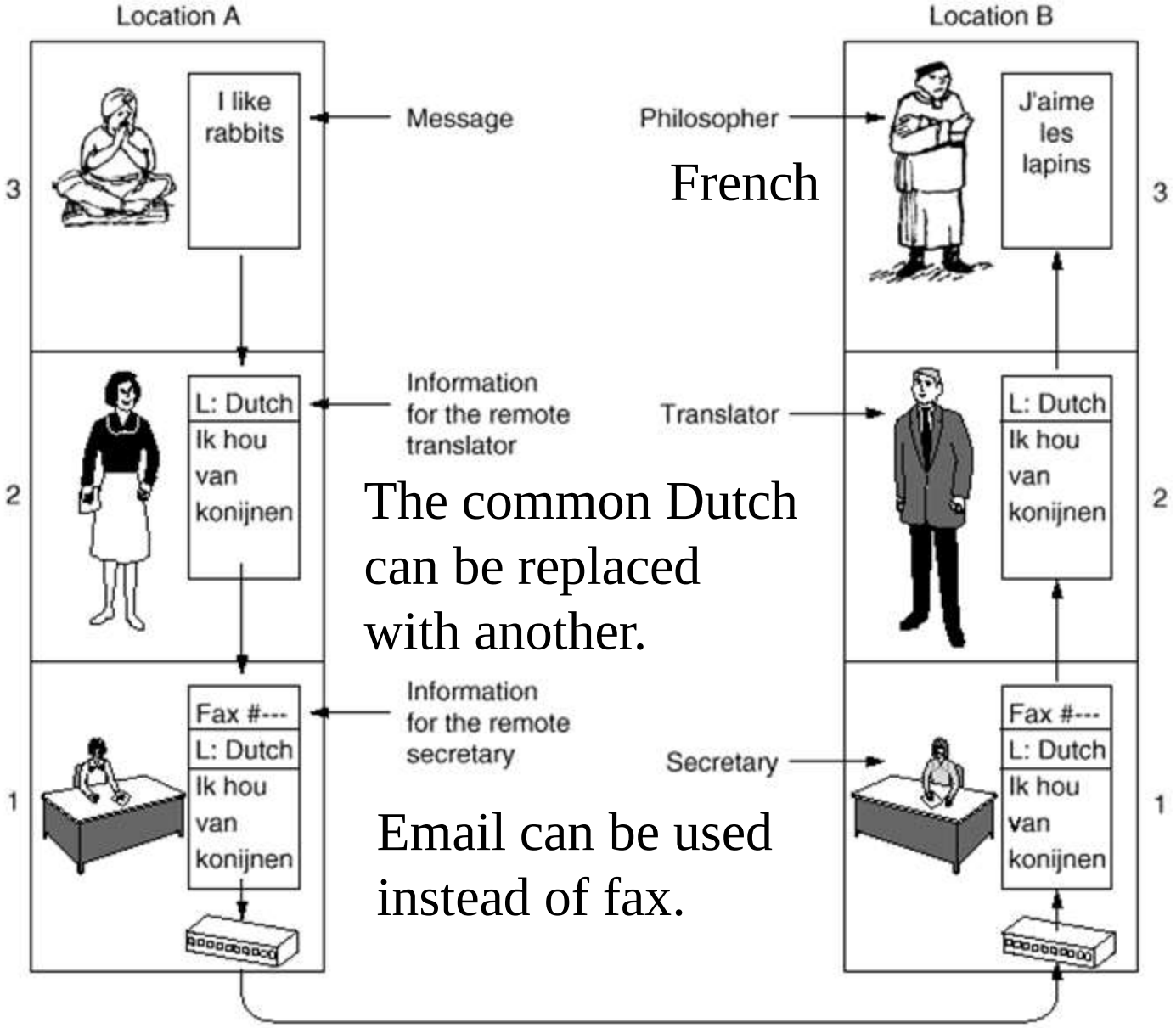
1.3 Network Software

1.3.1 Protocol Hierarchies

A set of layers and protocols is called a **network architecture**.

A list of protocols used by a certain system, one protocol per layer, is called a **protocol stack**.

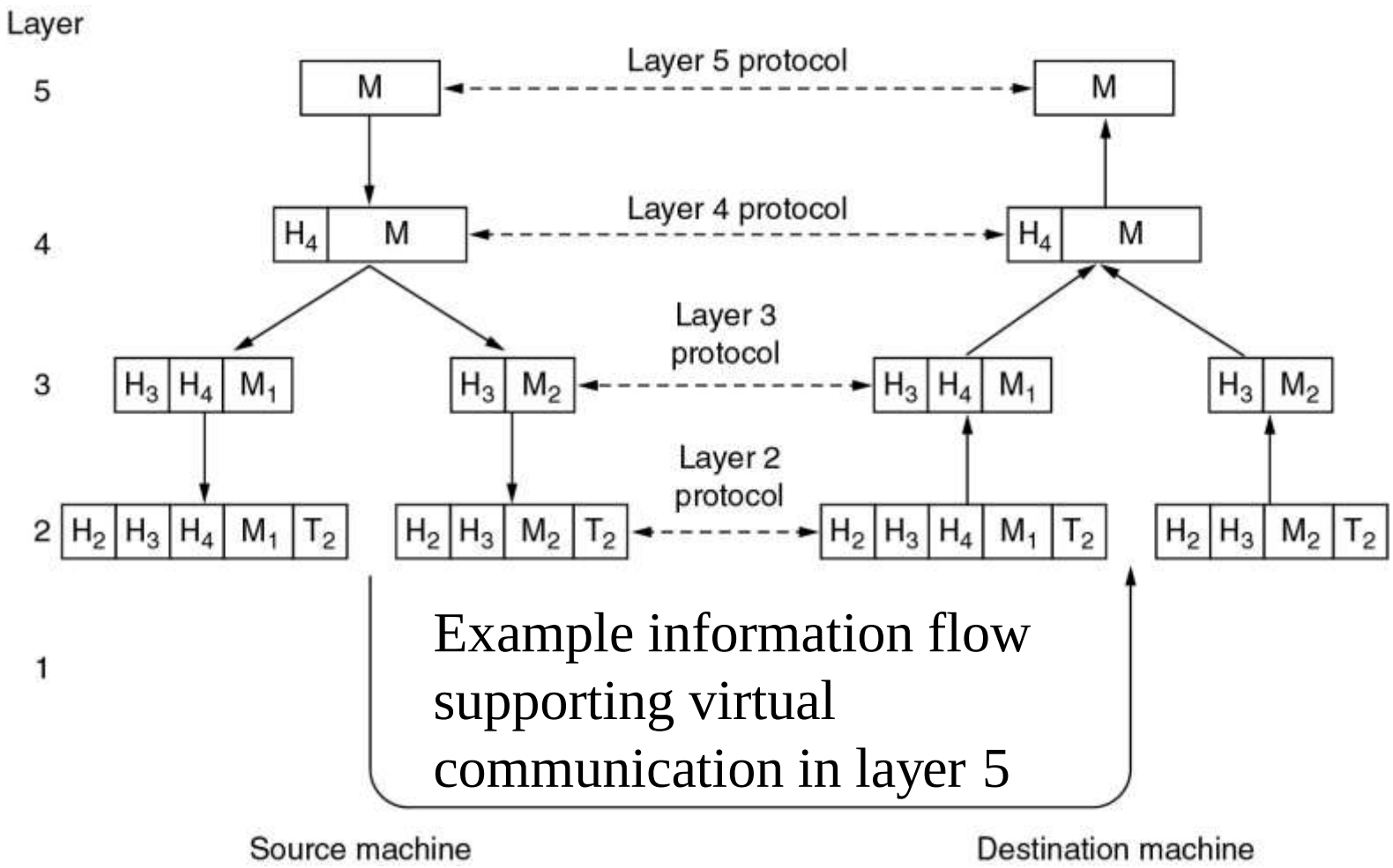
An analogy



The philosopher-translator-secretary architecture.

Chapter 1: Introduction

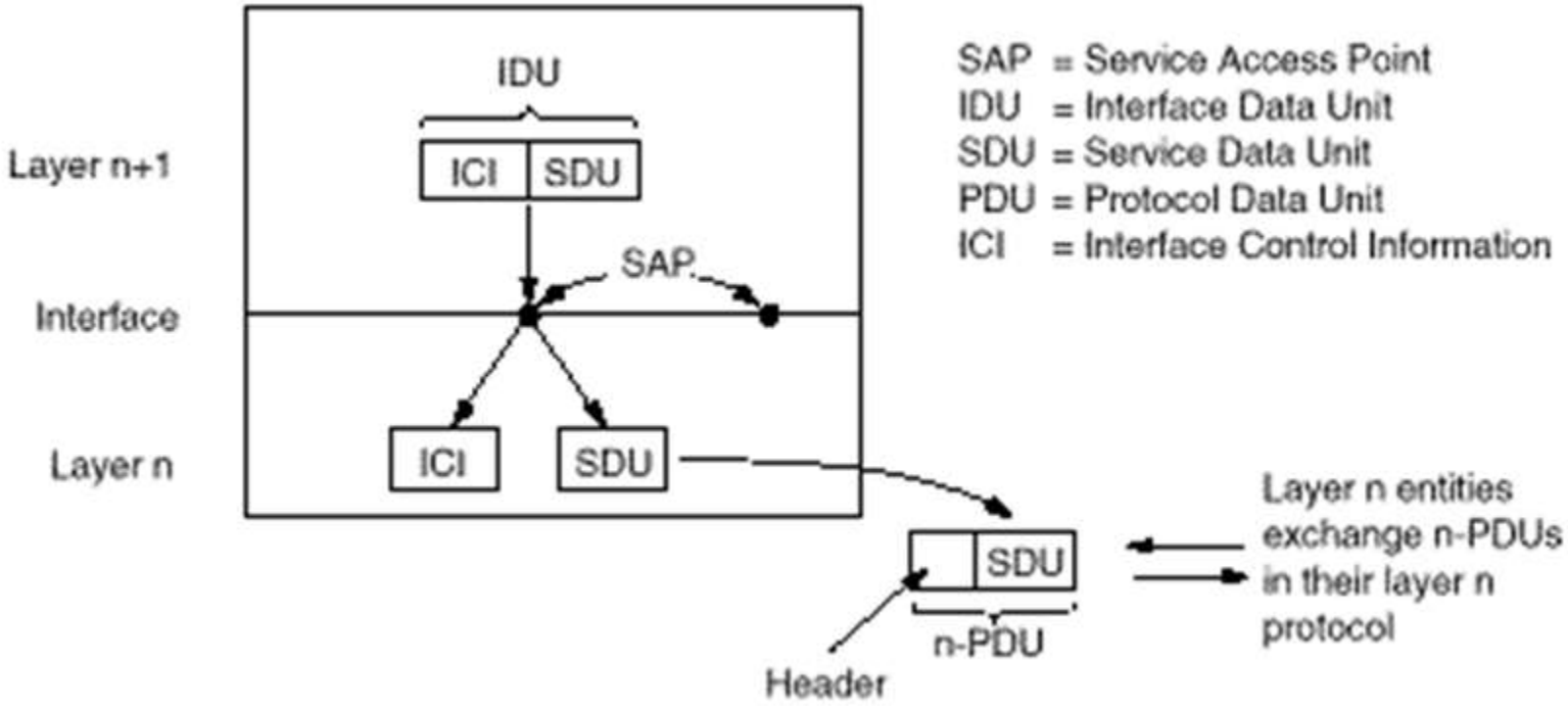
1.3 Network Software



Example information flow supporting virtual communication in layer 5

Chapter 1: Introduction

1.3 Network Software



Relations between layers at an interface

Chapter 1: Introduction

1.3 Network Software

1.3.2 Design Issues for the Layers

- A mechanism for identifying senders and receivers (naming and addressing)
- rules of transfer (simplex, half-duplex, full-duplex)
- error control (error correction and error detection)
- ordering and sequencing
- flow control, congestion control
- message or packet size (disassembling and reassembling)
- multiplexing and demultiplexing
- routing
- security

Chapter 1: Introduction

1.3 Network Software

1.3.3 Connection-Oriented and Connectionless Services

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Remote login
	Unreliable connection	Digitized voice
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Registered mail
	Request-reply	Database query

Six different types of service

Chapter 1: Introduction

1.3 Network Software

1.3.3 Connection-Oriented and Connectionless Services

Issue	Initial setup	Destination address	Packet sequence	Error control	Flow control	Option negotiation
Connection Oriented	required	only needed during setup	guaranteed	by network layer	by network layer	Yes
Connection-less	not possible	needed for every packet	not guaranteed	by transport layer	by transport layer	No

Chapter 1: Introduction

1.3 Network Software

1.3.4 Service Primitives

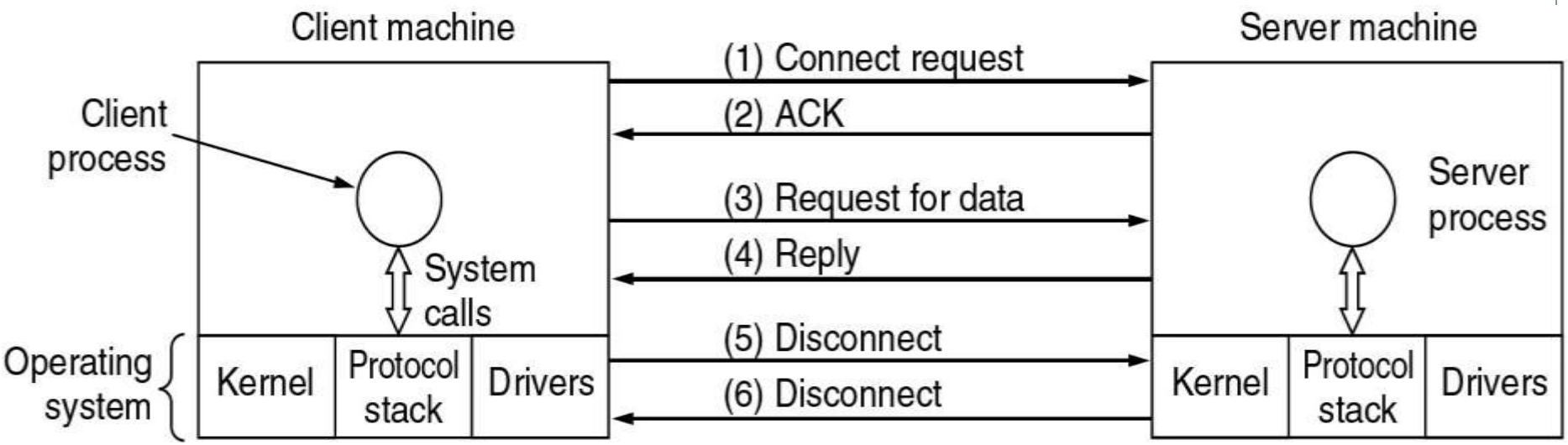
Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Five service primitives for implementing a simple connection-oriented service

Chapter 1: Introduction

1.3 Network Software

1.3.4 Service Primitives

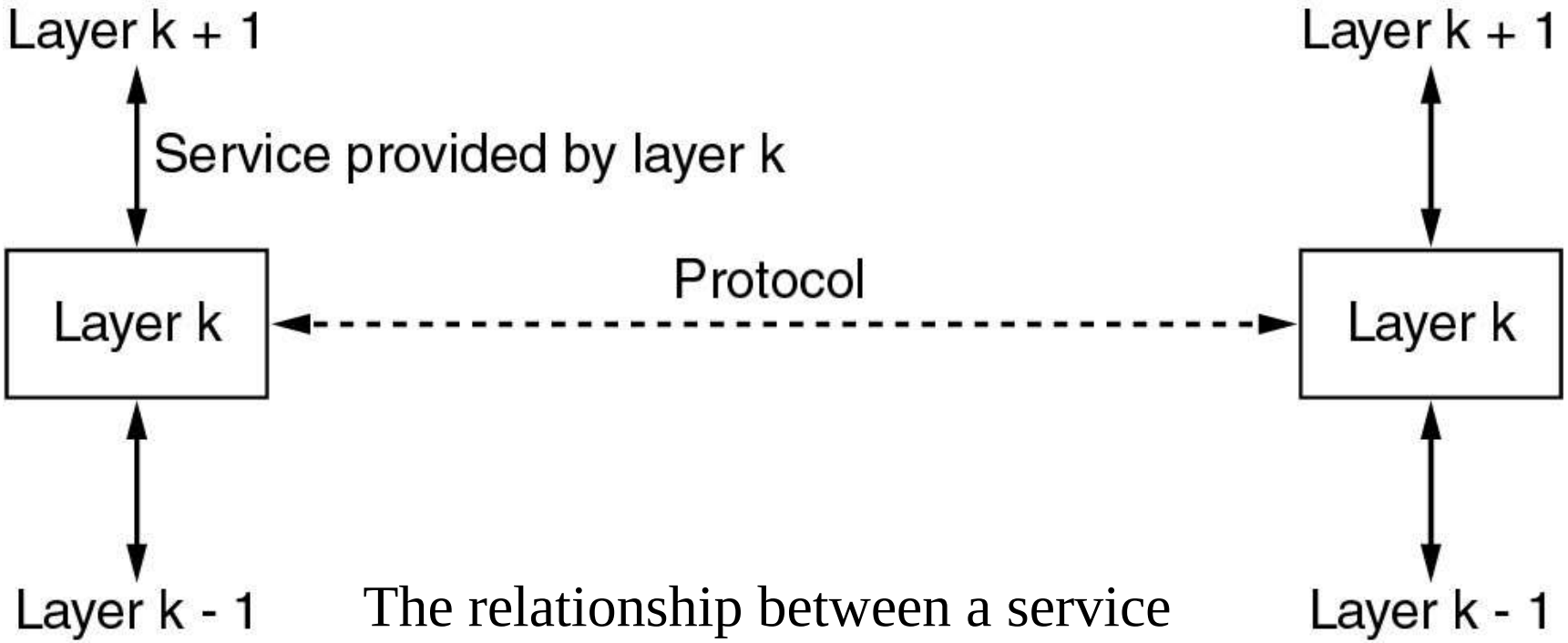


Packets sent in a simple client-server interaction on a connection-oriented network

Chapter 1: Introduction

1.3 Network Software

1.3.5 The Relationship of Services to Protocols



The relationship between a service and a protocol

Chapter 1: Introduction

1.3 Network Software

1.3.5 The Relationship of Services to Protocols

A service defines what operations the layer is prepared to perform on behalf of its users, but *it says nothing at all about how these operations are implemented*.

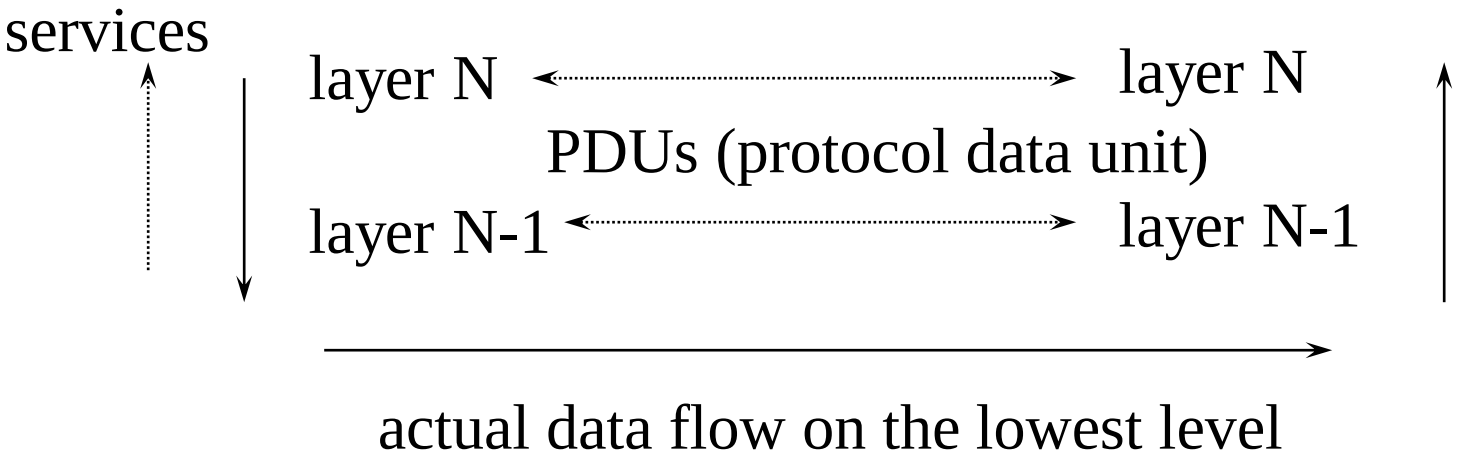
A protocol, in contrast, is a set of rules governing the format and meaning of the frames, packets, or messages that are exchanged by the peer entities within a layer. *Entities use protocols in order to implement their service definitions*.

Chapter 1: Introduction

1.4 Reference Models

1.4.1 The OSI Reference Model

In the late 1970s, to promote the **compatibility of network designs**, the *International Organization for Standardization* (ISO) proposed an architecture model called the *open systems interconnection references model* (OSI model).



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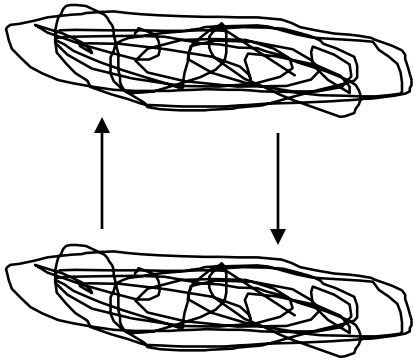
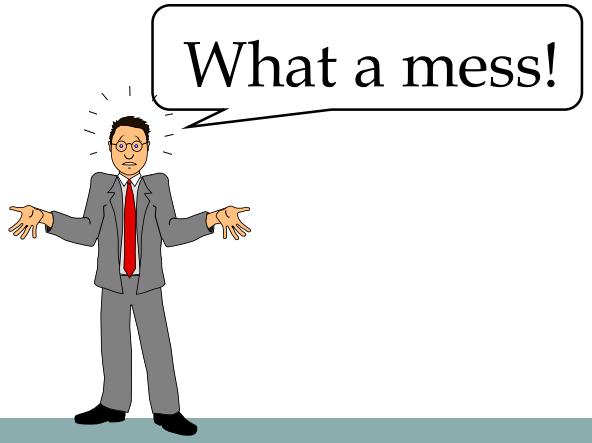
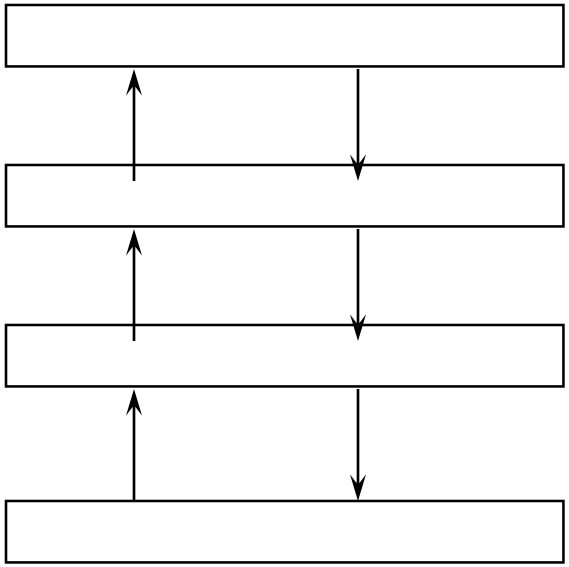
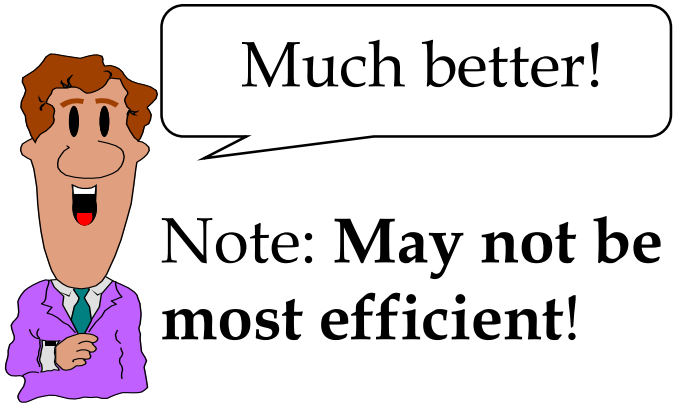
1.4 Reference Models

1.4.1 The OSI Reference Model

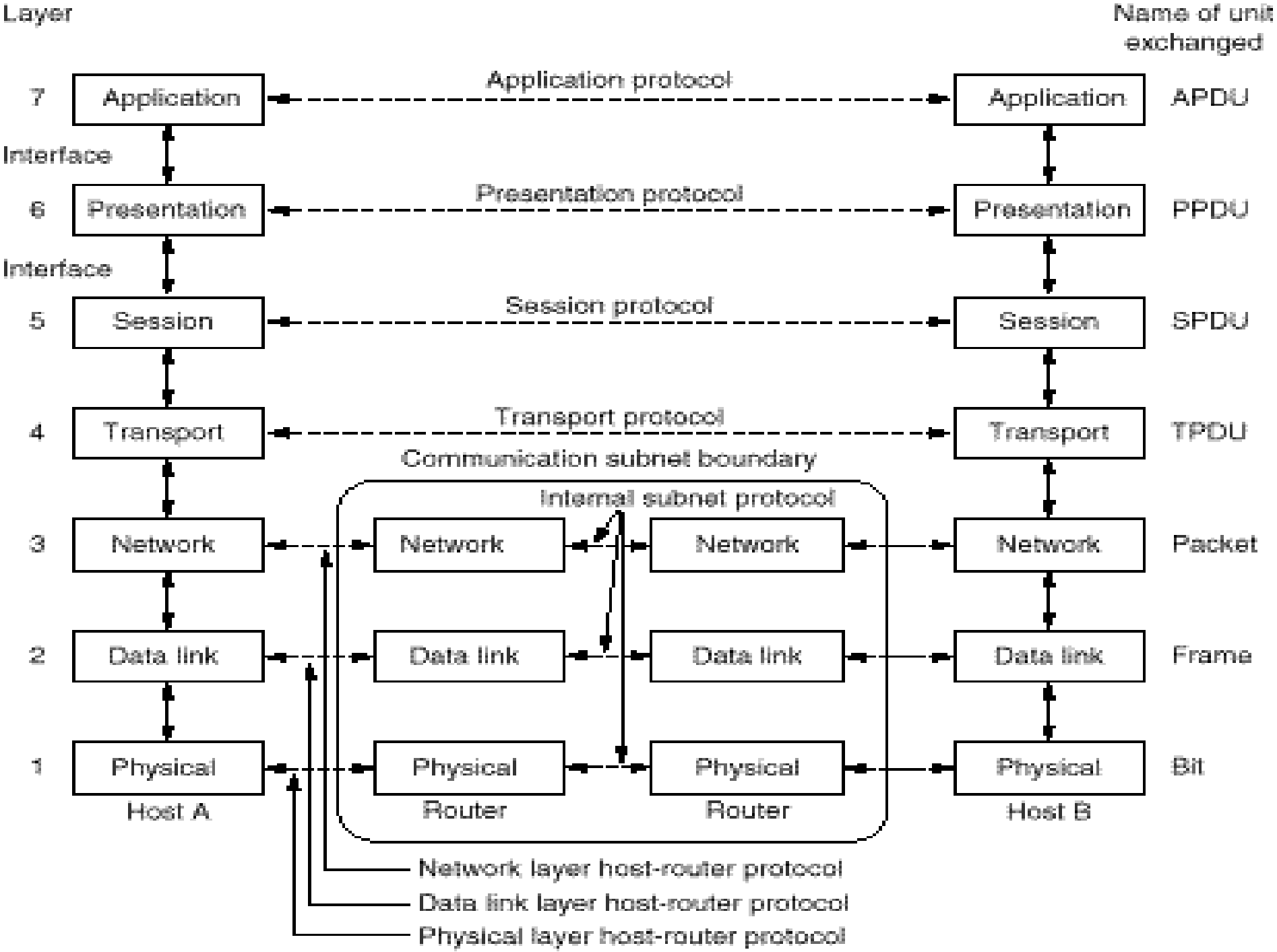








Note: **May not be the most efficient**!


Layer
Name of unit exchanged
7
Application
Application protocol
Application
APDU
Interface
6
Presentation
Presentation protocol
Presentation
PPDU
Interface
5
Session
Session protocol
Session
SPDU
4
Transport
Transport protocol
Transport
TPDU
Communication subnet boundary
Internal subnet protocol
3
Network
Network
Network
Network
Packet
2
Data link
Data link
Data link
Data link
Frame
1
Physical
Physical
Physical
Physical
Bit
Host A
Router
Router
Host B
Network layer host-router protocol
Data link layer host-router protocol
Physical layer host-router protocol

Chapter 1: Introduction

1.4 Reference Models

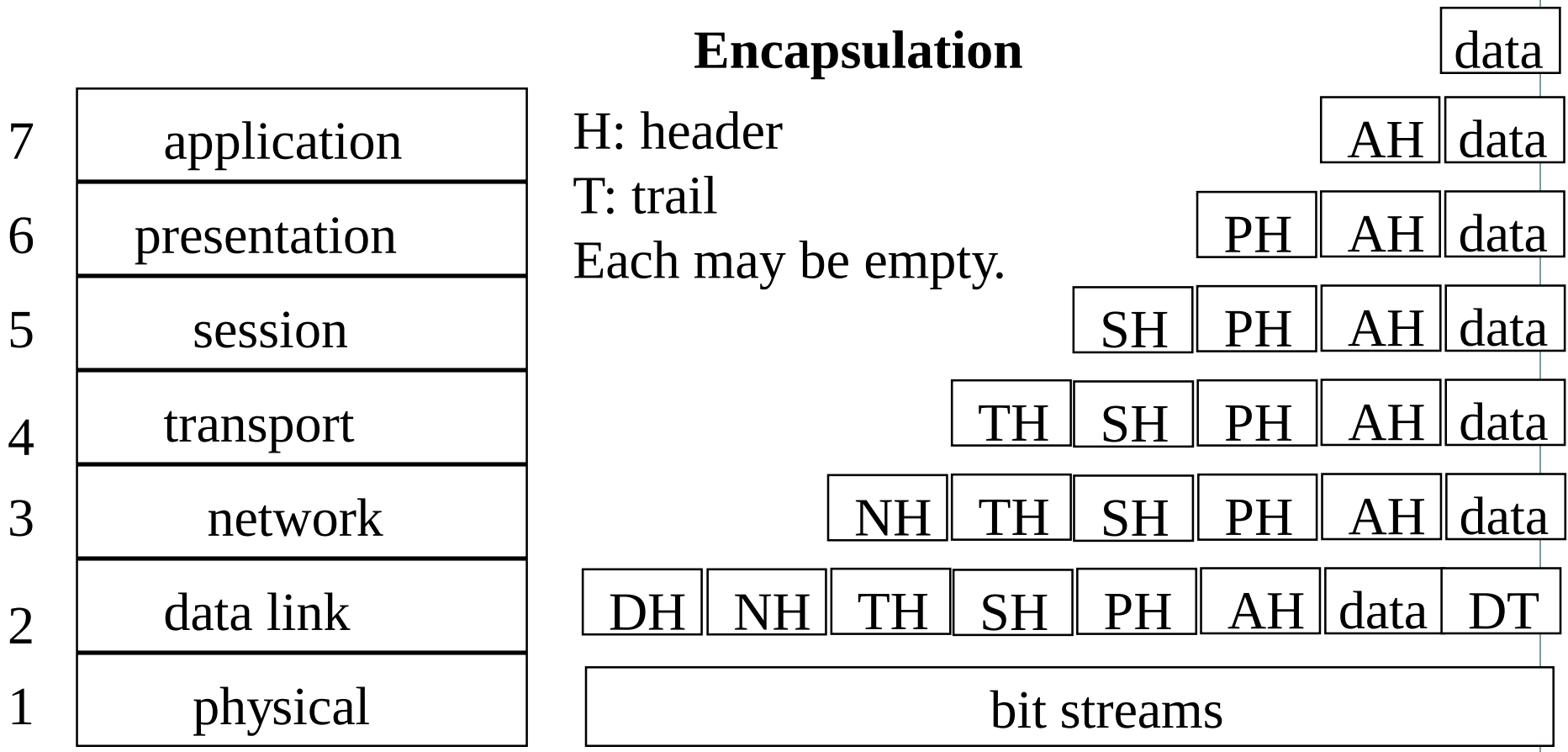
1.4.1 The OSI Reference Model

7	application	network services (email, file transfer)
6	presentation	formatting, encryption, and compression
5	session	setup and management of end-to-end conversation
4	transport	end-to-end delivery of messages
3	network	end-to-end transmission of packets
2	data link	transmission of packets on one given link
1	physical	transmission of bits

Chapter 1: Introduction

1.4 Reference Models

1.4.1 The OSI Reference Model

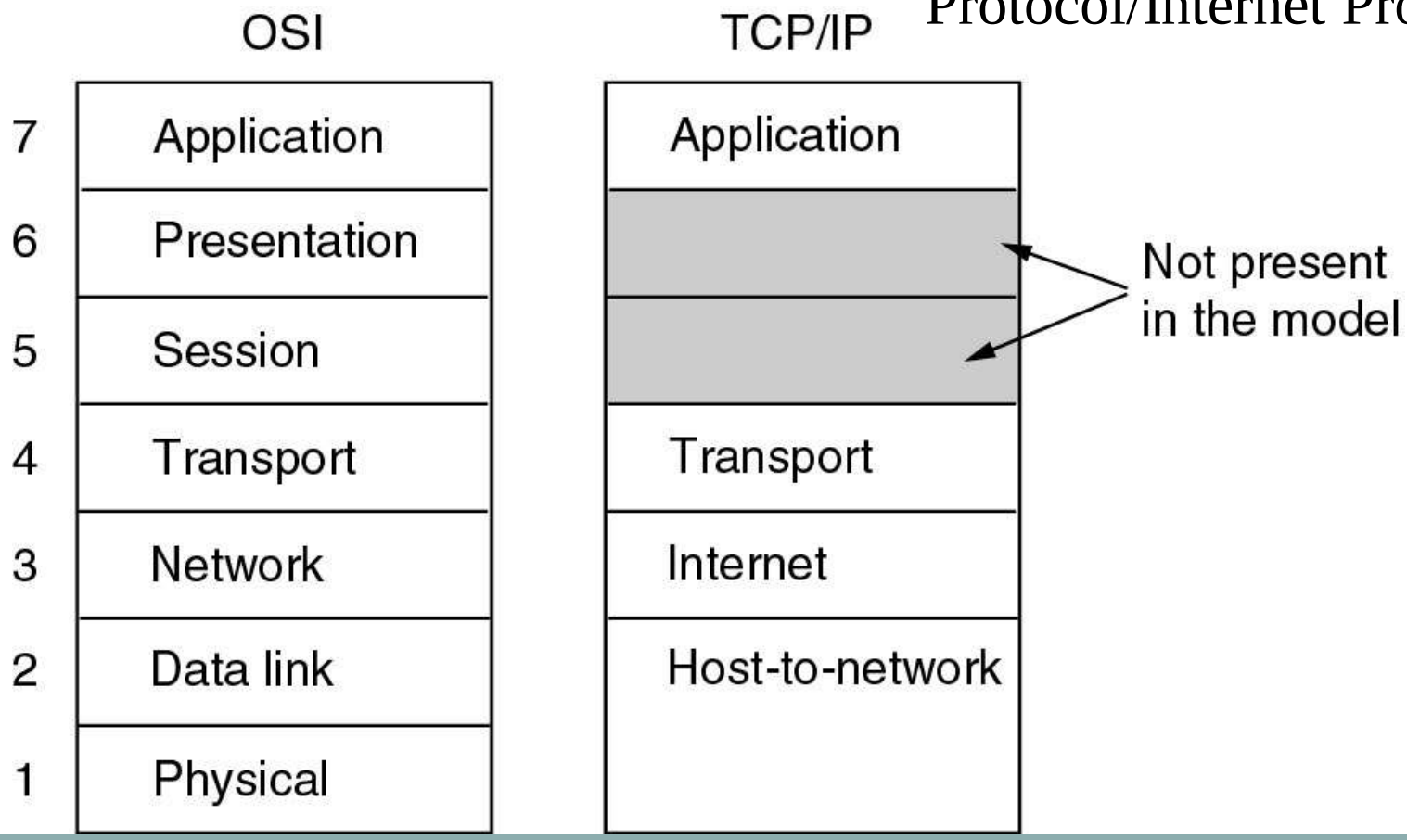


Chapter 1: Introduction

1.4 Reference Models

1.4.2 The TCP/IP Reference Model

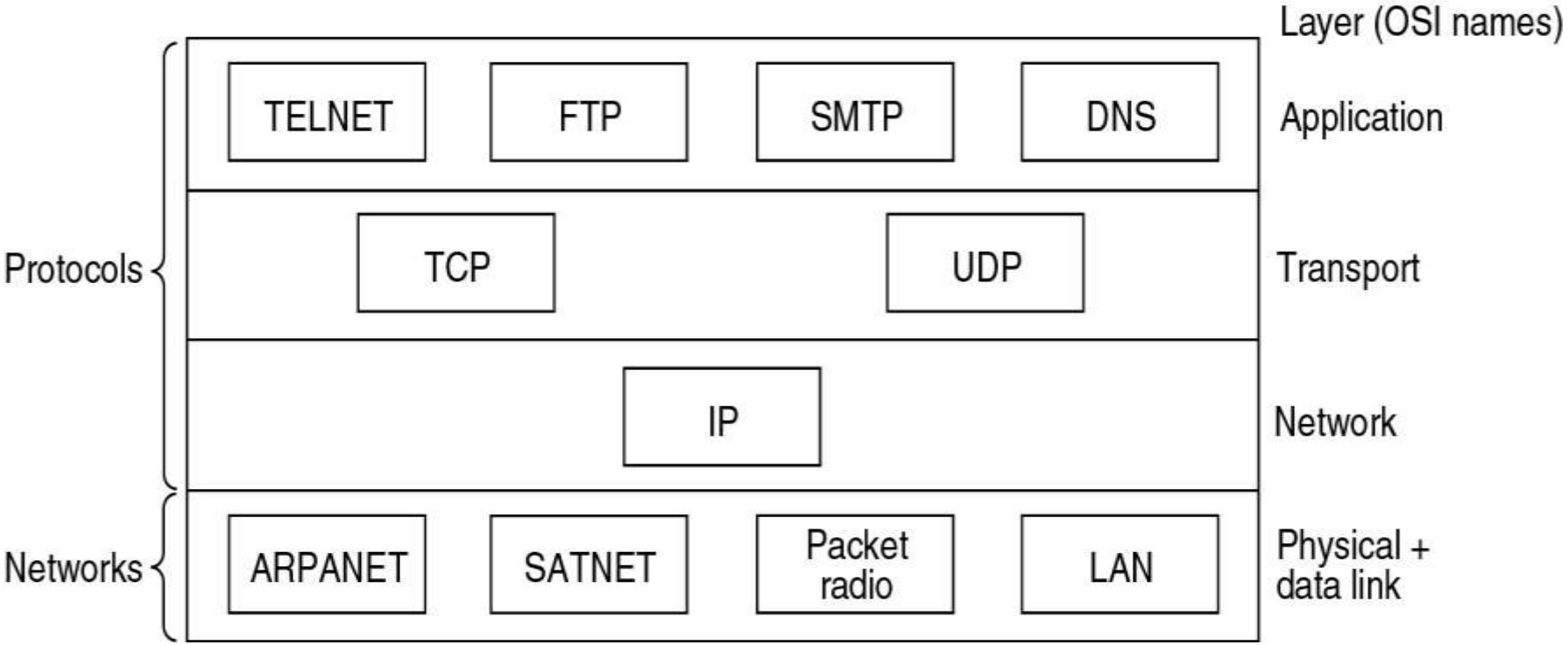
(Transmission Control Protocol/Internet Protocol



Chapter 1: Introduction

1.4 Reference Models

1.4.2 The TCP/IP Reference Model



Protocols and networks in the TCP/IP model initially

Chapter 1: Introduction

1.4 Reference Models

1.4.3 A Comparison of the OSI and TCP/IP Reference Model

The OSI reference model was devised before the (OSI) protocols were invented.

- Not biased toward any one particular set of protocols
- The designers did not have much experience with the subject and did mot have a good idea of which functionality to put in which layer

Chapter 1: Introduction

1.4 Reference Models

1.4.3 A Comparison of the OSI and TCP/IP Reference Model

With the TCP/IP, the protocols came first, and the model was really just a description of the existing protocols. *The model did not fit any other protocol stacks*.

7 layers versus 4
connection-oriented versus connectionless

Chapter 1: Introduction

1.4 Reference Models

1.4.4 A Critique of the OSI Model and Protocols

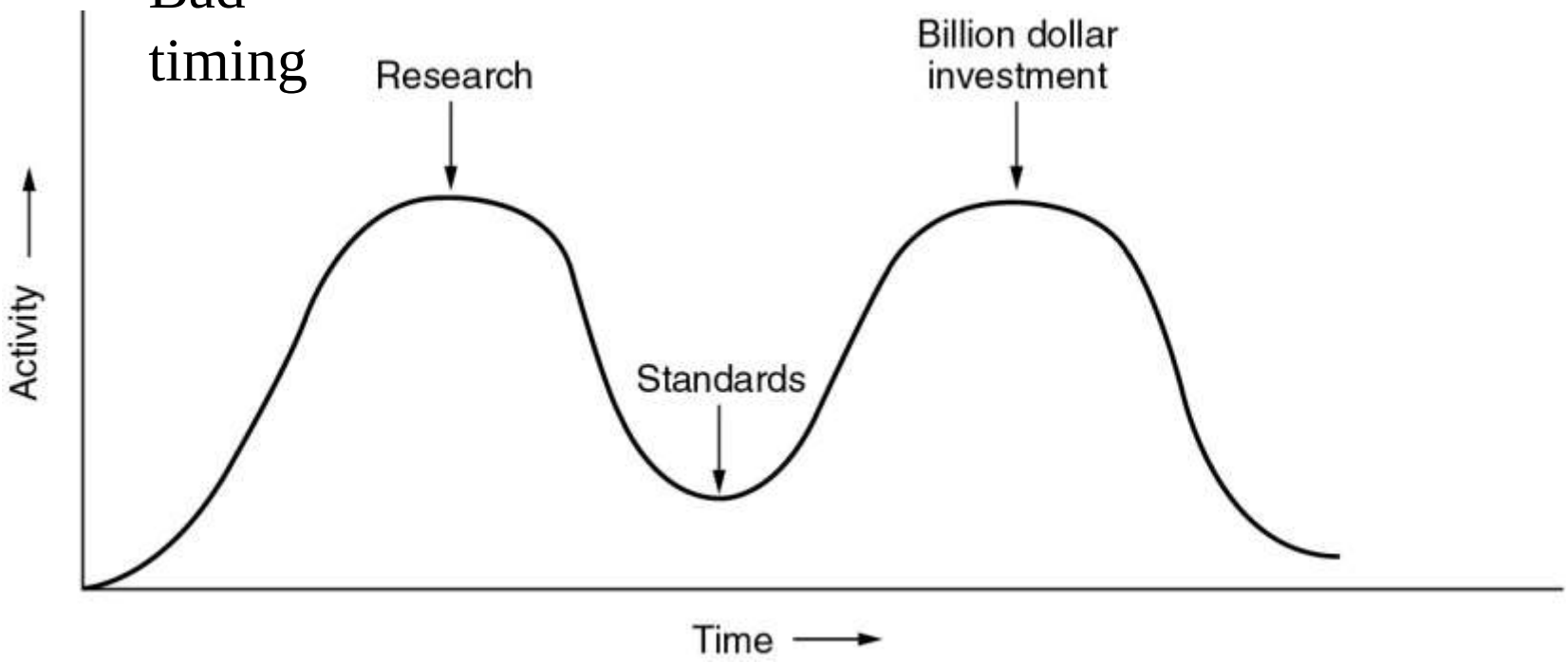
1. Bad timing
2. Bad technology
3. Bad implementation
4. Bad politics

Chapter 1: Introduction

1.4 Reference Models

1.4.4 A Critique of the OSI Model and Protocols

Bad timing



The apocalypse of the two elephants

Chapter 1: Introduction

1.4 Reference Models

1.4.5 A Critique of the TCP/IP Reference Model

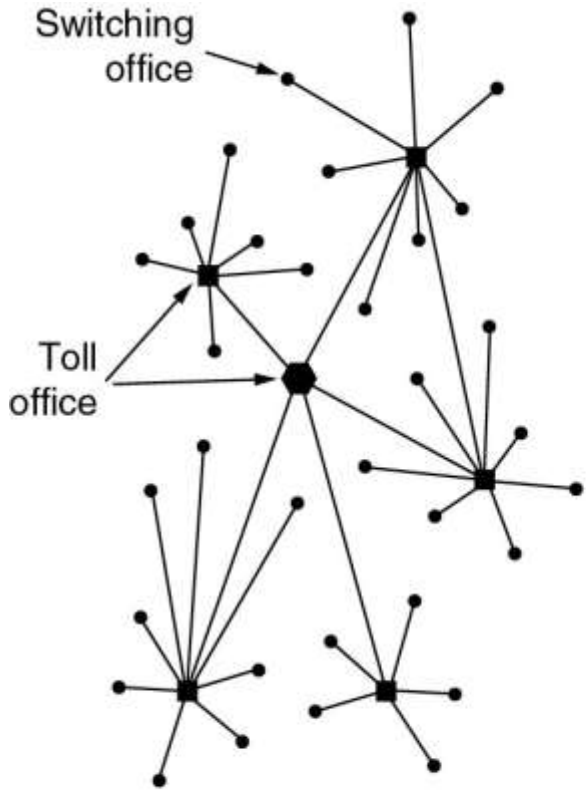
1. The model does not clearly distinguish the concepts of service, interface, and protocol.
2. The model is not general and is poorly suited to describing and other protocol stack.
3. The model does not distinguish (or even mention) the physical and data link layer.
4. Only TCP and IP are carefully thought out and implemented. Many other protocols are ad hoc.

Chapter 1: Introduction

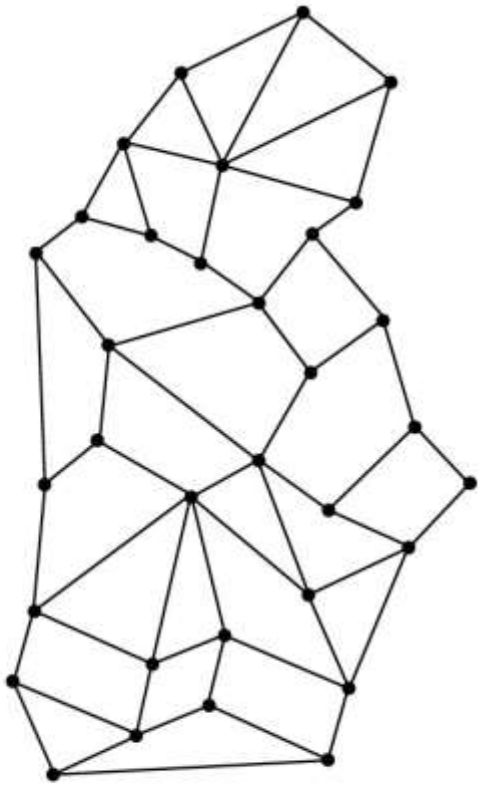
1.5 Example Networks

1.5.1 The Internet

(a) Structure of the telephone system.
(b) Baran's proposed distributed switching system



(a)



(b)

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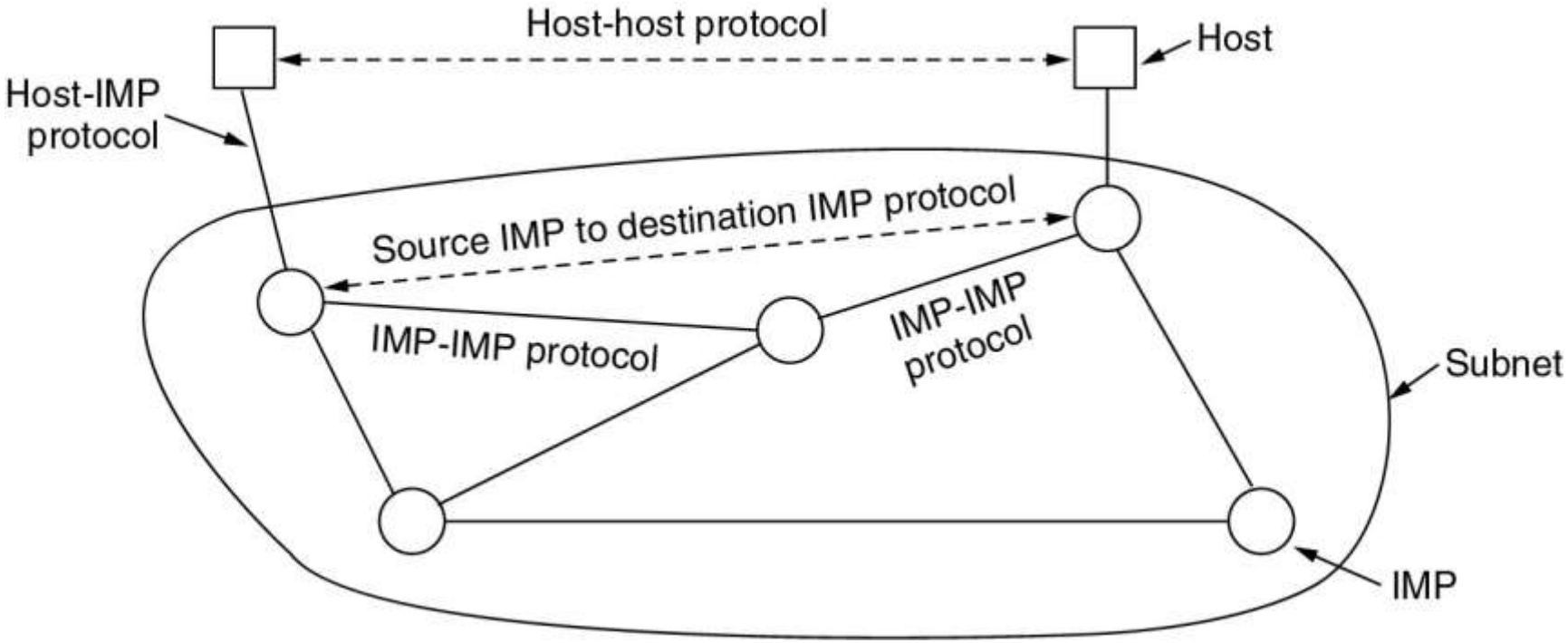
1.5 Example Networks

The ARPANET

IMP: Interface Message Processor



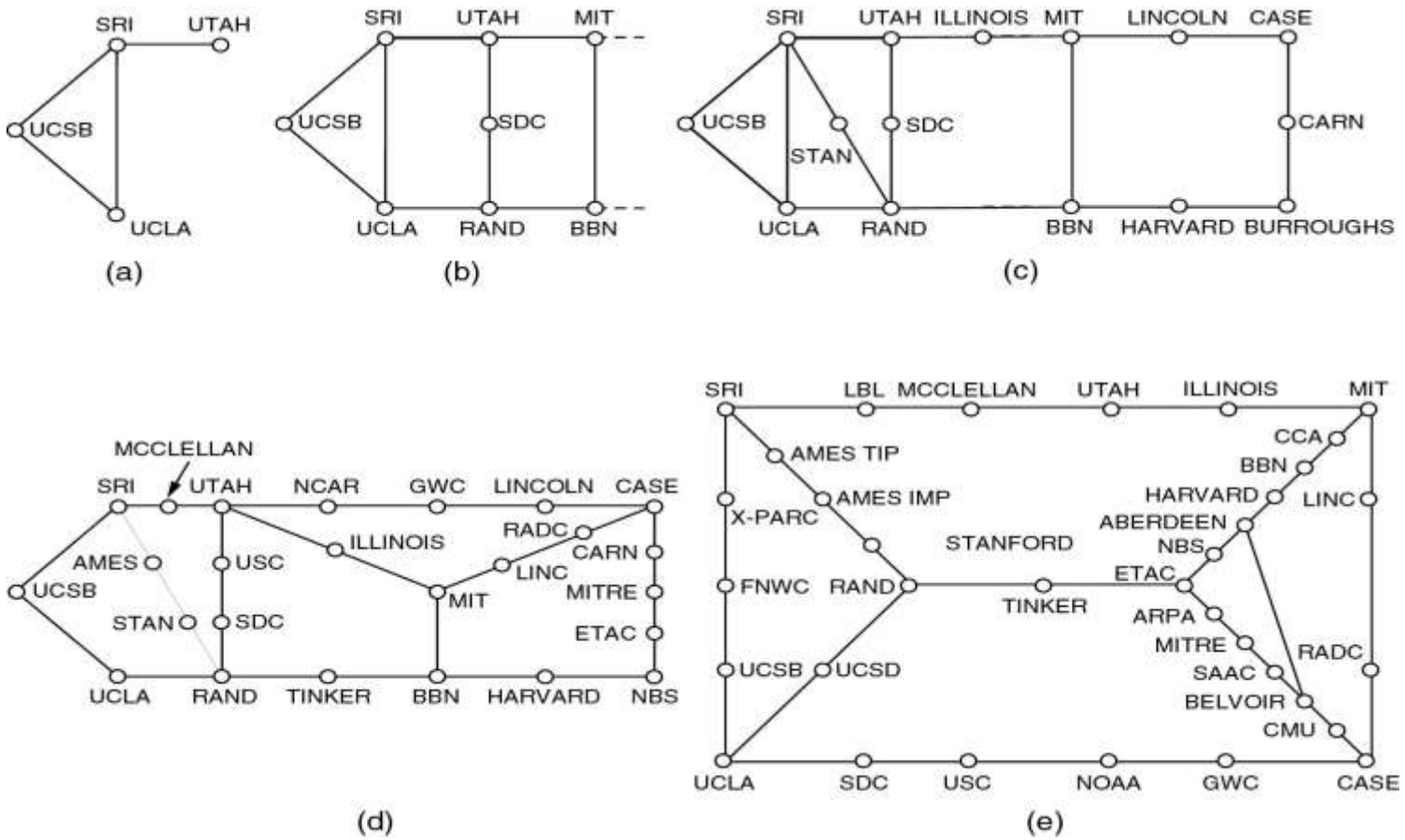
The original ARPANET design

Chapter 1: Introduction

1.5 Example Networks

1.5.1 The Internet

Growth of the ARPANET (a) December 1969. (b) July 1970. (c) March 1971. (d) April 1972. (e) September 1972



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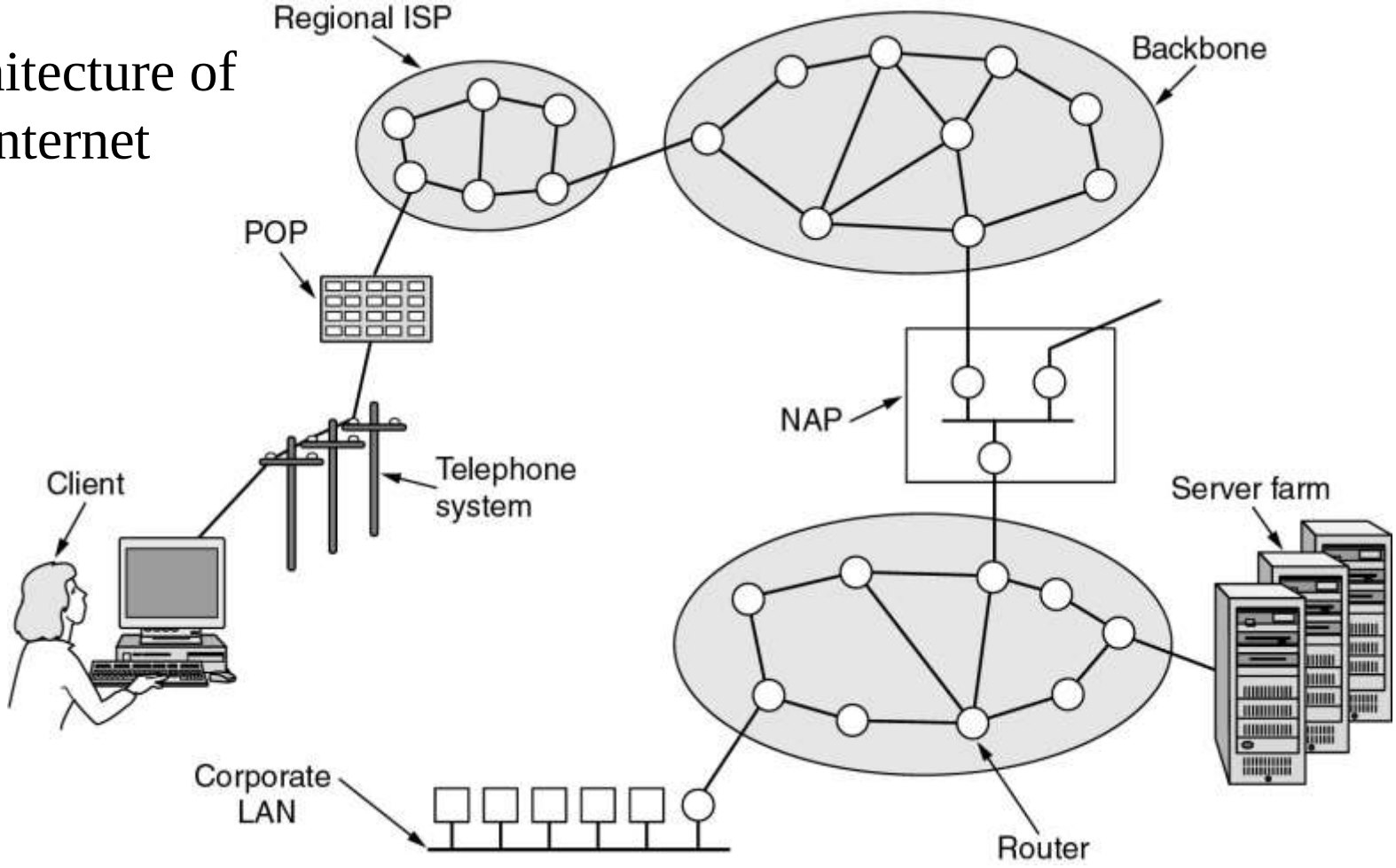
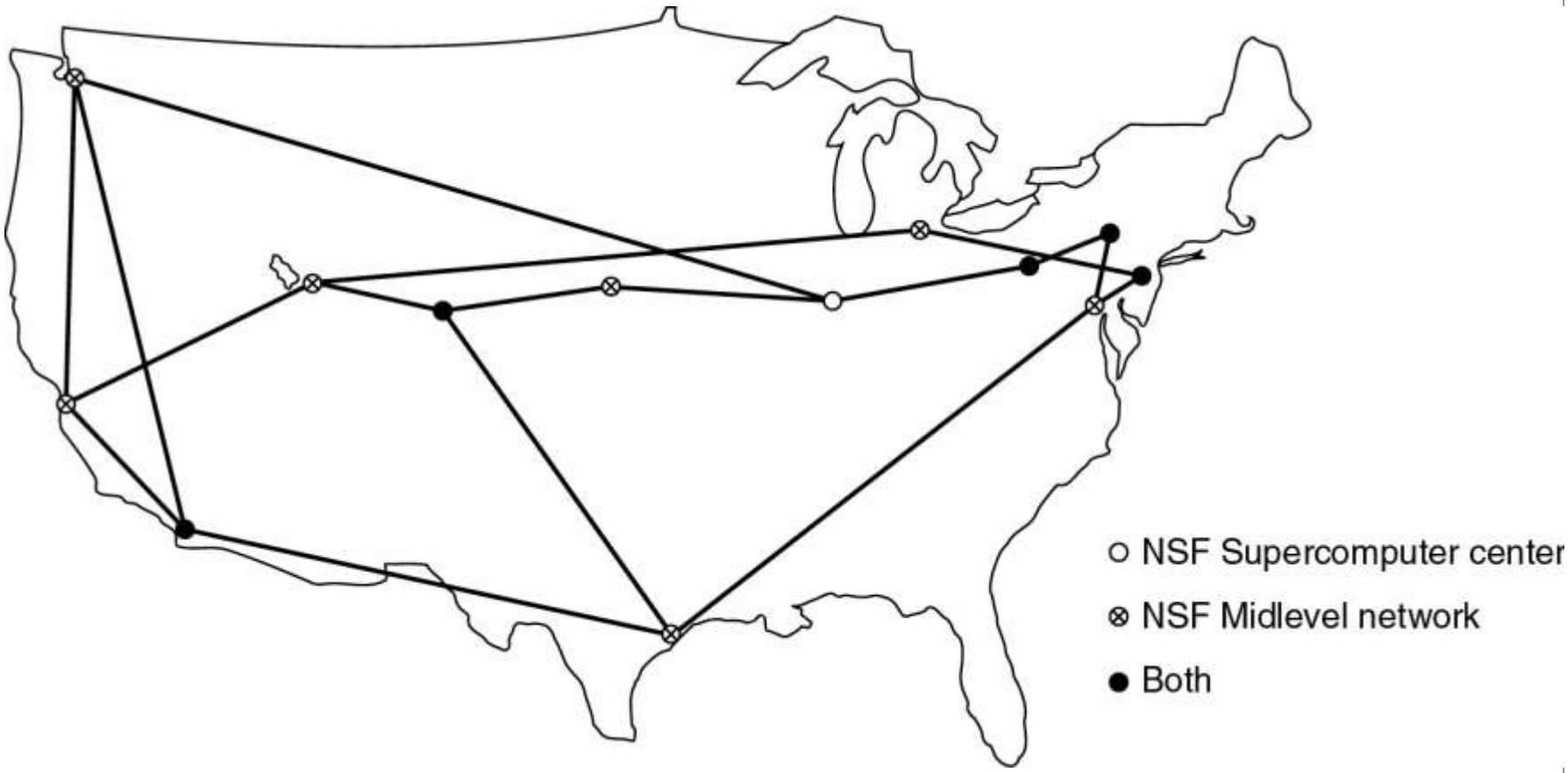
1.5 Example Networks

Architecture of the Internet



Chapter 1: Introduction

1.5 Example Networks



The NSFNET backbone in 1988

Chapter 1: Introduction

1.5 Example Networks

Internet Usage

Traditional applications (1970 – 1990)

1. E-mail
2. News
3. Remote login
4. File transfer

Now: ?

Chapter 1: Introduction

1.5 Example Networks

The Internet Society http://www.isoc.org

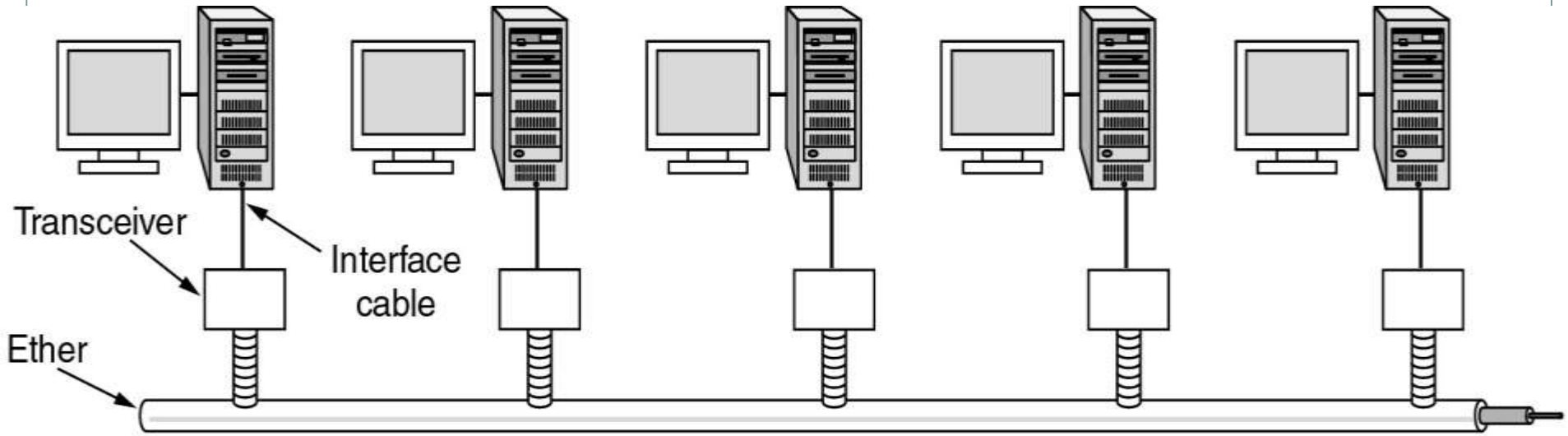
A brief history of the Internet
http://www.isoc.org/internet/history/brief.html

Hobbes' Internet Timeline
http://www.isoc.org/guest/zakon/Internet/History/HIT.html

Chapter 1: Introduction

1.5 Example Networks

Ethernet

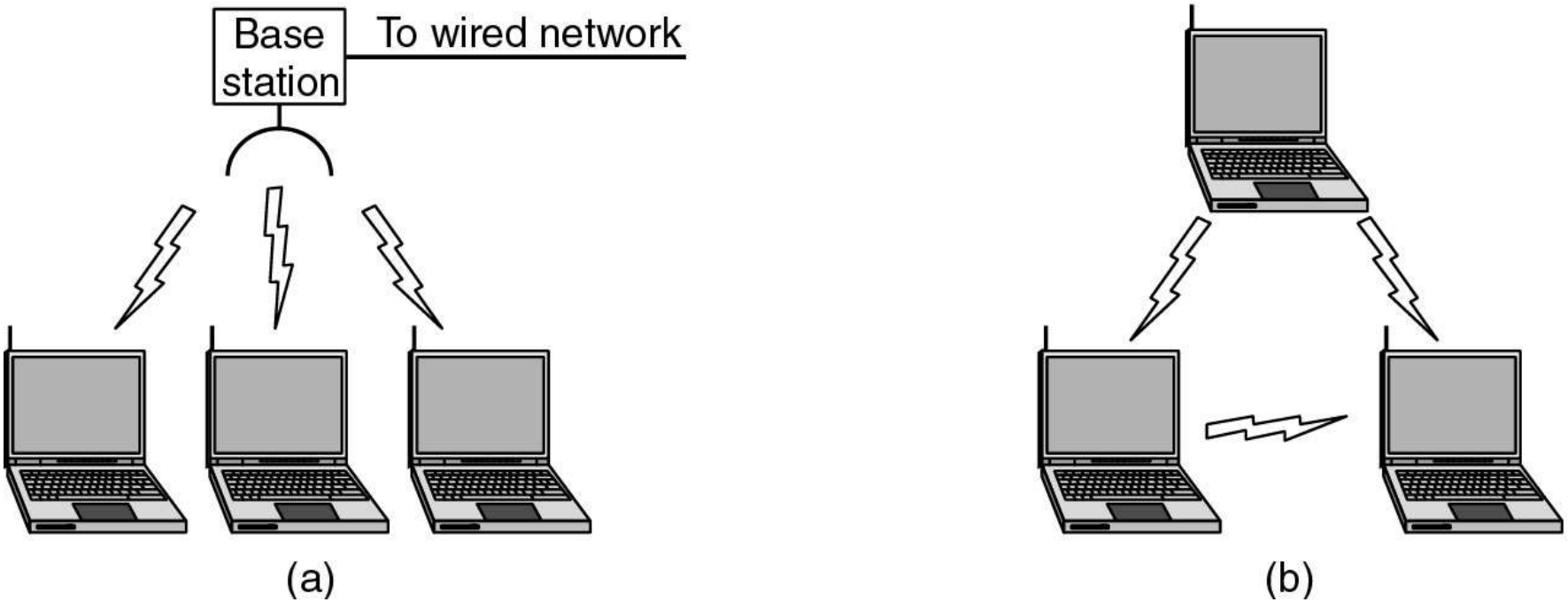


Architecture of the original Ethernet

Chapter 1: Introduction

1.5 Example Networks

Wireless LANs

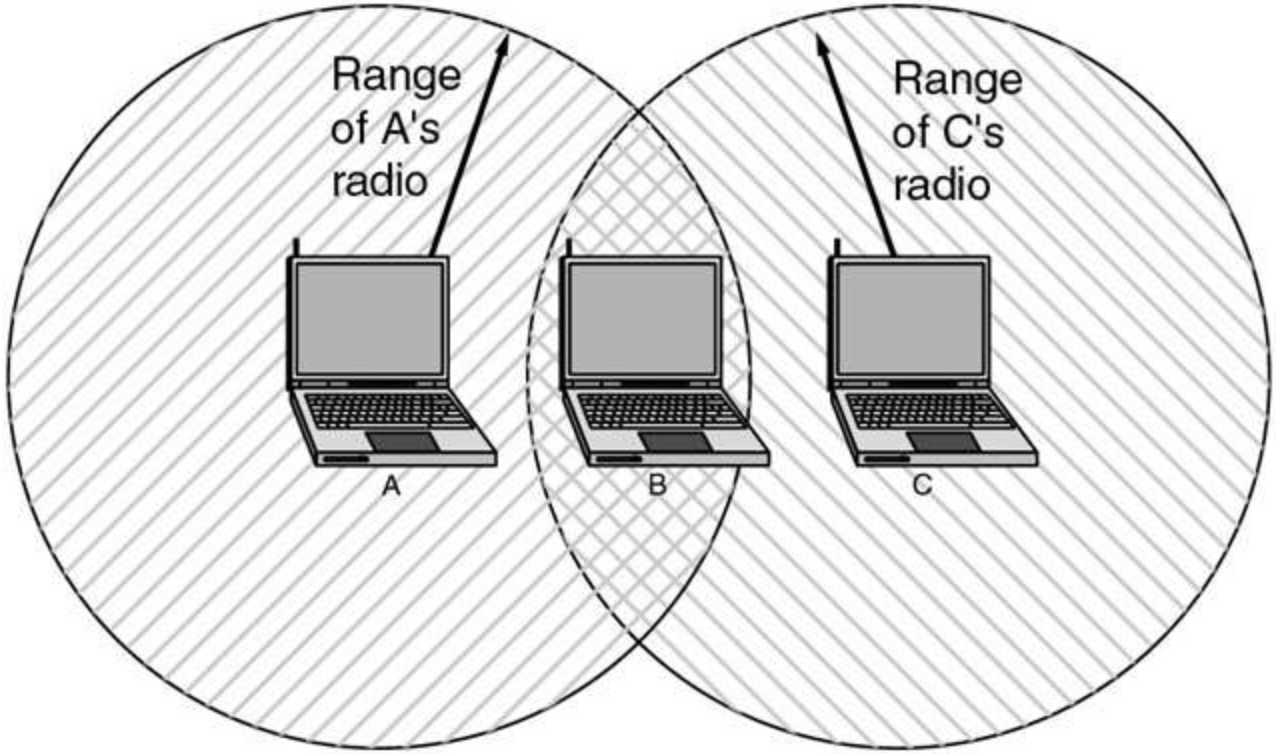


(a) Wireless networking with a base station.
(b) Ad hoc networking

Chapter 1: Introduction

1.5 Example Networks

Wireless LANs



The range of a single radio may not cover the entire system

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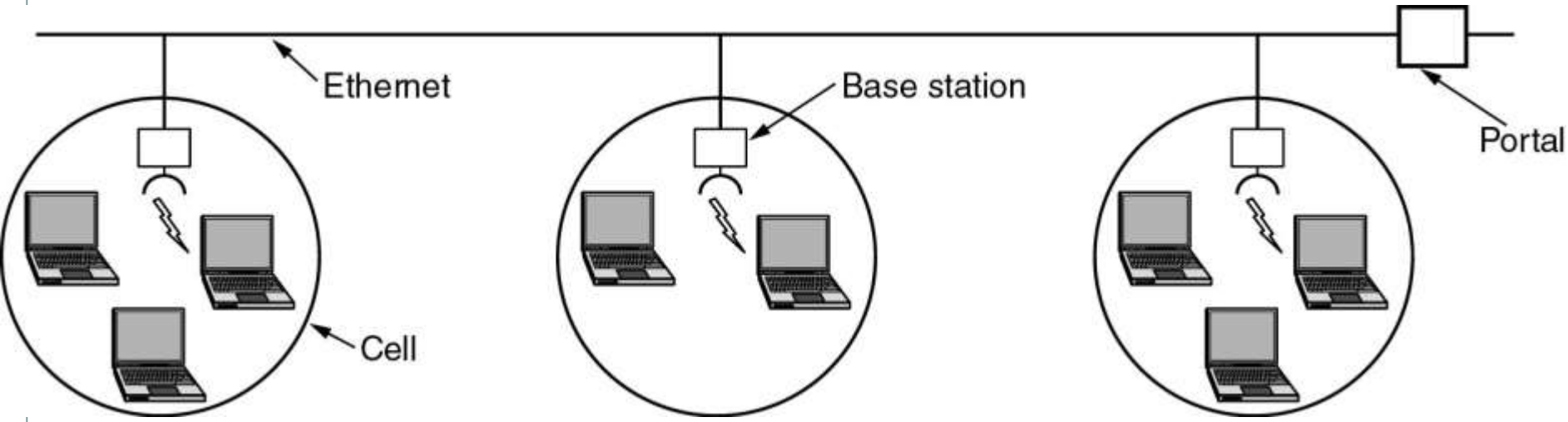
1.5 Example Networks

Wireless LANs



A multicell 802.11 network

Chapter 1: Introduction

1.6 Network Standardization

Benefits of standards

De facto (from the fact)
De jure (by law)

standard

Chapter 1: Introduction

1.6 Network Standardization

1.6.1 Who's Who in the Telecommunications World

AT&T

NOKIA
CONNECTING PEOPLE

NTT

FCC

Sony Ericsson

MOTOROLA

中華電信

MCI WORLDCOM

EIA
Electronic Industries Alliance

FAR EASTONE
遠傳

國家通訊傳播委員會
NATIONAL COMMUNICATIONS COMMISSION

台湾大哥大

Chapter 1: Introduction

1.6 Network Standardization

1.6.2 Who's Who in the International Standards World

International Telecommunication Union

ORGANISATION INTERNATIONALE DE NORMALISATION

ISO

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ETSI World Class Standards

ANSI Online American National Standards Institute

ANSI Online is designed to provide you

IEEE

NIST National Institute of Standards and Technology

. . .working with industry to develop and apply technology, measurements and standards

財団法人 日本規格協会

Japanese Standards Association

Chapter 1: Introduction

1.6 Network Standardization

1.6.2 Who's Who in the International Standards World

ITU

- Main sectors
  - Radiocommunications
  - Telecommunications Standardization
  - Development
- Classes of Members
  - National governments
  - Sector members
  - Associate members
  - Regulatory agencies

Chapter 1: Introduction

1.6 Network Standardization

1.6.2 Who's Who in the International Standards World

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number. Nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth)
802.16 *	Broadband wireless
802.17	Resilient packet ring

Chapter 1: Introduction

1.6 Network Standardization

1.6.3 Who's Who in the Internet Standards World

IETF

Internet Society

iana
Internet Assigned Numbers Authority

ICANN

Chapter 1: Introduction

1.7 Metric Units

Exp.	Explicit	Prefix	Exp.	Explicit	Prefix
10^{-3}	0.001	milli	10^{3}	1,000	Kilo
10^{-6}	0.000001	micro	10^{6}	1,000,000	Mega
10^{-9}	0.000000001	nano	10^{9}	1,000,000,000	Giga
10^{-12}	0.000000000001	pico	10^{12}	1,000,000,000,000	Tera
10^{-15}	0.000000000000001	femto	10^{15}	1,000,000,000,000,000	Peta
10^{-18}	0.000000000000000001	atto	10^{18}	1,000,000,000,000,000,000	Exa
10^{-21}	0.000000000000000000001	zepto	10^{21}	1,000,000,000,000,000,000,000	Zetta
10^{-24}	0.000000000000000000000001	yocto	10^{24}	1,000,000,000,000,000,000,000,000	Yotta

The principal metric prefixes

Chapter 1: Introduction

Exercises:

1. Do a research on the Internet service providers in Taiwan. Compare their costs and access methods.

Page 81, problem 3.
Page 82, problem 14.
Page 82, problem 20.
Page 83, problem 28.
Page 83, problem 35.